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1.Project Topic:

The topic of this project is the development of **Smart Market** – a smart online store management system that combines e-commerce functionalities with artificial intelligence (AI) in order to improve the customer experience and optimize store operations.

The system is designed to serve three main types of users: **Customers**, **Workers**, and **Managers**, where each user type has a dedicated interface and specific permissions according to their role.

The main uniqueness of the system is the integration of **AI** as a support tool for customers. Customers can search for products using natural language, not only by exact product names. In addition, on each product page, the customer can ask the AI questions about the product itself, such as:

- How to use the product,
- How often it should be used,
- Who the product is suitable for,
- What the main benefits or features of the product are.

With these capabilities, the customer can receive guidance and information independently, without the need for external assistance or customer service support. This improves the shopping experience, saves time, and helps customers make more informed purchasing decisions.

In addition, the system provides:

- A complete online shopping environment including product browsing, shopping cart, orders, and payments.
- Internal management tools for order picking, order tracking, and employee management.
- Administrative tools for product management, customer request management, and statistical reports.

The project combines the worlds of e-commerce and intelligent systems in order to create a modern, efficient, and user-friendly platform for both customers and store staff.

2.Detailed Description of the Project:

The **Smart Market** project is a web-based system for managing a smart online store, combining e-commerce functionalities with artificial intelligence (AI) in order to improve user experience and optimize store operations.

The system is divided into three main parts according to the different user roles: **Customer Side**, **Worker Side**, and **Manager Side**.

2.1 Customer Side

The customer side is intended for end users of the store and allows them to perform the entire shopping process online and independently. Users can register to the system or log in to an existing account, and then browse products and categories.

The system allows:

- Regular product search by text, as well as smart search using AI.
- Viewing products by categories and accessing a detailed product page.
- On the product page, the user can ask the AI questions about the product itself, such as:
  how to use the product, how often it should be used, who the product is suitable for, and what its main benefits are.
  This feature enables customers to get guidance and information independently, without the need for external assistance or customer support.
- Adding products to the shopping cart and to the favorites list.
- Placing orders and completing payments through the system.
- Viewing previous orders and tracking their status.
- Receiving notifications about order status changes (ready for pickup, delivered, etc.).
- Sending requests to customer support using the “Contact Us” form.

2.2 Worker Side

The worker side is designed for store employees who are responsible for preparing orders. Workers log into the system through a dedicated login page (Worker Login) and get access to orders that are ready for picking.

Using the system, workers can:

- View orders filtered by status (for example, orders that have been paid and are waiting for picking).
- Select an order and open its picking screen.
- View the list of products in the order, including image, name, and quantity.
- Mark each product after it has been picked.
- Confirm the completion of the picking process, after which the order status is updated to **ready_for_pickup**.

This process helps ensure accurate order preparation and reduces mistakes during fulfillment.

2.3 Manager Side

The manager side includes all the capabilities of the worker side, and in addition provides advanced management tools for controlling the store.

The manager can:

- Manage products: add new products, update existing products, and delete products.
- Manage employees: add new workers to the system and remove existing workers.
- Manage orders: view and monitor all orders in the system across all statuses (to be picked, ready for pickup, delivered).
- Manage customer requests: view requests sent by customers, add internal notes, and mark requests as handled.
- View statistics and reports: including sales data, top-selling products, and revenue over time, in order to support business decision-making.

Through these tools, the manager gains full control over the store’s operational and business activities.

2.4 Integration of AI in the System

One of the key components of the project is the integration of artificial intelligence. The AI is used in two main areas:

- Smart product search based on free-text user queries.
- Answering user questions on the product page, based on the product data stored in the system.

This integration allows customers to receive accurate and relevant information, improves the overall user experience, and reduces the workload on customer support.

3.System Objectives

The main objective of the **Smart Market** system is to provide a smart, efficient, and user-friendly platform for managing an online store, combining e-commerce functionalities with artificial intelligence (AI) tools.

The specific objectives of the system are:

1. **Improve Customer Experience**
   To enable customers to search for products easily, receive intelligent recommendations, and get clear answers about products using AI, without the need for external assistance.
2. **Support Independent Decision-Making**
   To allow customers to ask questions directly on the product page (such as how to use a product or how often to use it) and receive helpful information, helping them make better purchasing decisions.
3. **Provide a Complete Online Shopping Process**
   To support the full shopping flow, including registration, login, product browsing, favorites, shopping cart, payment, order tracking, and notifications.
4. **Optimize Order Fulfillment Operations**
   To provide workers with tools for efficient order picking, including viewing orders, picking products, and updating order statuses accurately.
5. **Enable Effective Store Management**
   To provide managers with tools to manage products, employees, orders, and customer requests from a single system.
6. **Support Data-Driven Decision Making**
   To present managers with statistics and reports about sales, top-selling products, revenue trends, and customer requests.
7. **Ensure Role-Based Access Control**
   To provide different interfaces and permissions for customers, workers, and managers, ensuring that each user can access only the functions relevant to their role.
8. **Improve Efficiency and Reduce Errors**
   To reduce manual mistakes in order preparation and management by using structured workflows and status-based order tracking.

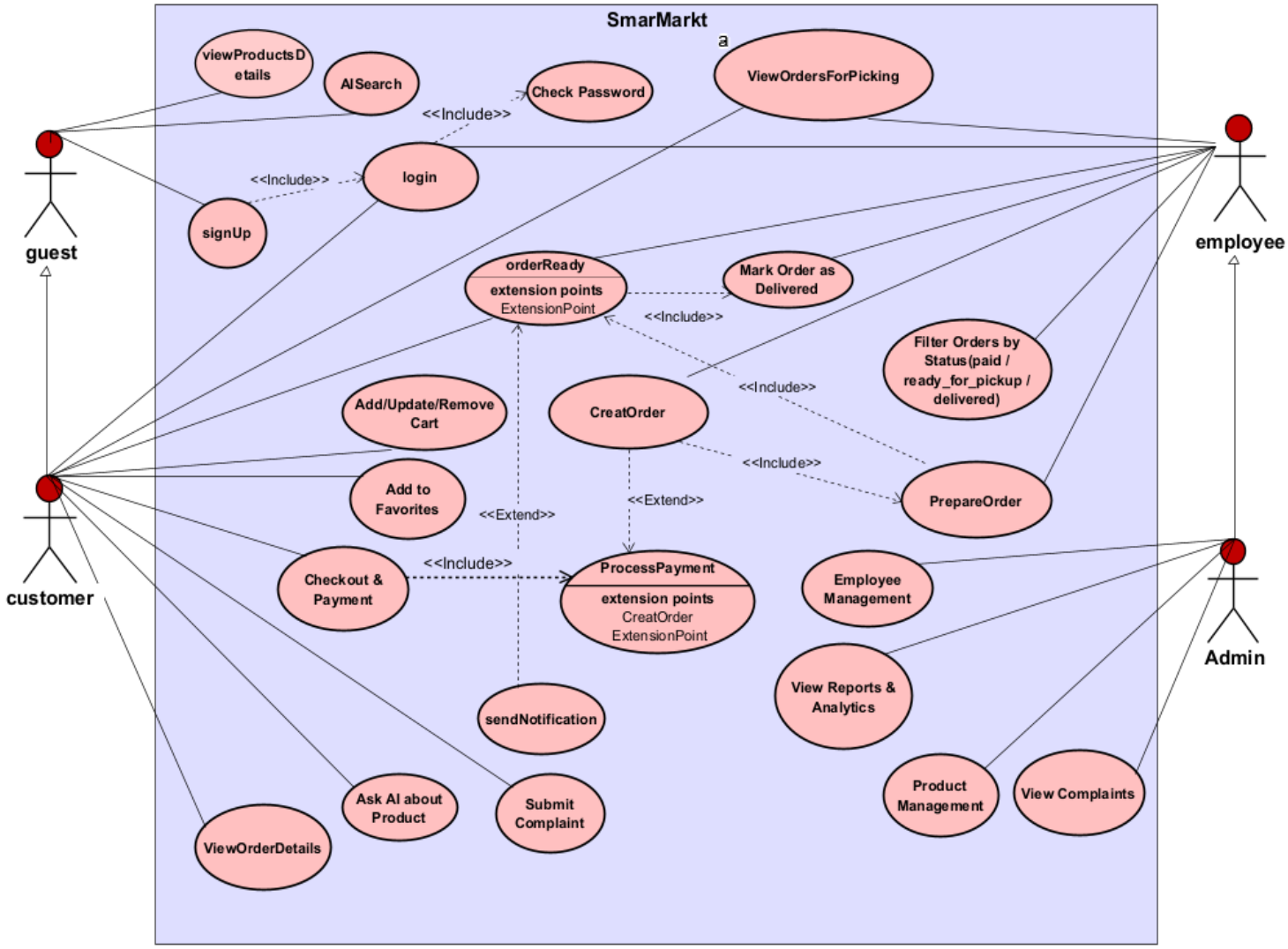
4.1 Functional Requirements:

	Requirements
1.	The system shall allow users to register and create a new account.
2.	The system shall allow users to log in using email and password.
3.	The system shall allow customers to search for products using text and AI-based search.
4.	The system shall allow customers to view products by categories.
5.	The system shall allow customers to view detailed product pages.
6.	The system shall allow customers to ask the AI questions about a specific product.
7.	The system shall allow customers to add and remove products from favorites.
8.	The system shall allow customers to add and remove products from the shopping cart.
9.	The system shall allow customers to place orders and make payments.
10.	The system shall allow customers to view their order history and order status.
11.	The system shall send notifications to customers about order status changes.
12.	The system shall allow customers to send requests using the Contact Us form.
13.	The system shall allow workers to view orders that are ready for picking.
14.	The system shall allow workers to pick products from an order and mark them as picked.
15.	The system shall update the order status to ready_for_pickup after picking is completed.
16.	The system shall allow managers to manage products (add, update, delete).
17.	The system shall allow managers to manage employees (add, delete).
18.	The system shall allow managers to view and manage customer requests.
19.	The system shall allow managers to view statistics and reports.
20.	The system shall support role-based access control (Customer, Worker, Manager).

4.2Non-Functional Requirements:

	Requirements
1.	The system shall provide secure authentication and password protection.
2.	The system shall be user-friendly and easy to use.
3.	The system shall respond to user actions within a reasonable time.
4.	The system shall protect user data and payment information.
5.	The system shall be available through a web browser.

5.Use Case Diagram – SmartMarket:



Use Case Diagram Description – SmartMarket

Use Case: View Products
Actor: Guest / Customer

Actor Actions	System Responses
The user opens the SmartMarket website	The system loads the homepage and displays main categories
The user selects a product category	The system displays all products in the selected category
The user clicks on a product	The system displays the product details (name, price, description)

Use Case: Sign Up
Actor: Guest

Actor Actions	System Responses
The guest clicks on "Sign Up"	The system displays the registration form
The guest enters personal information	The system validates the entered data
The guest submits the form	The system creates a new account and shows a success message

Use Case: AI Search
Actor: Guest / Customer

Actor Actions	System Responses
The user opens the SmartMarket website	The system displays the search bar on the homepage
The user enters a search query (e.g., product name or description)	The system analyzes the query using the AI search engine
The user clicks on "Search"	The system displays a list of relevant products
The user selects a product from the results	The system displays the product details page

Use Case: Login

Actor: Guest / Customer

Actor Actions	System Responses
The user enters email and password	The system verifies the credentials
The user clicks "Login"	The system grants access and opens the user dashboard
The user enters wrong credentials	The system displays an error message

Use Case: Add / Update / Remove Cart

Actor: Customer

Actor Actions	System Responses
The customer clicks "Add to Cart"	The system adds the product to the shopping cart
The customer changes product quantity	The system updates the cart total price
The customer removes a product	The system removes the product from the cart

Use Case: Checkout & Payment

Actor: Customer

Actor Actions	System Responses
The customer clicks "Proceed to Checkout"	The system displays the checkout page with total cost
The customer enters payment details	The system validates the payment information
The customer confirms payment	The system processes the payment and shows confirmation
The payment is successful	The system creates the order and sends a notification

Use Case: Add to Favorites

Actor: Customer

Actor Actions	System Responses
The customer views a product	The system displays the product details page
The customer clicks "Add to Favorites"	The system adds the product to the favorites list
The customer opens the favorites list	The system displays all saved favorite products

Use Case: View Order Details

Actor: Customer

Actor Actions	System Responses
The customer opens the "My Orders" page	The system displays a list of the customer's orders
The customer selects an order	The system displays detailed information about the order
The customer views the order status	The system shows the current status of the order

Use Case: Ask AI about Product

Actor: Customer

Actor Actions	System Responses
The customer opens a product page	The system displays the "Ask AI" option
The customer types a question about the product	The system processes the question using the AI assistant
The customer submits the question	The system displays the AI-generated answer

Use Case: Submit Complaint

Actor: Customer

Actor Actions	System Responses
The customer opens the "Submit Complaint" page	The system displays the complaint form
The customer fills in the complaint details	The system validates the input
The customer submits the complaint	The system saves the complaint and shows a confirmation message

Use Case: Send Notification

Actor: System (Triggered by Customer / Employee actions)

Actor Actions	System Responses
The customer completes a payment	The system creates a notification for order confirmation
The employee updates the order status	The system sends a notification to the customer
The system triggers a notification event	The system delivers the notification by email or in-app message
The customer opens the notification	The system displays the notification details

Use Case: View Orders for Picking

Actor: Employee / Admin

Actor Actions	System Responses
The employee opens the orders page	The system displays a list of pending orders
The employee filters orders by status	The system shows orders matching the selected status
The employee selects an order	The system displays the order details

Use Case: Filter Orders by Status

Actor: Employee / Admin

Actor Actions	System Responses
The employee opens the orders list	The system displays all orders
The employee selects a status filter (e.g., Paid, Ready, Delivered)	The system filters and displays matching orders
The employee changes the filter	The system updates the displayed list

Use Case: View Orders for Picking

Actor: Employee / Admin

Actor Actions	System Responses
The employee opens the orders management page	The system displays a list of orders to be prepared
The employee selects an order	The system displays the order details
The employee starts preparing the order	The system marks the order as "In Preparation"

Use Case: Prepare Order

Actor: Employee / Admin

Actor Actions	System Responses
The employee selects an order to prepare	The system marks the order as "In Preparation"
The employee finishes preparing the order	The system updates the order status to "Ready for Pickup"
The system updates the status	The system sends a notification to the customer

Use Case: Create Order (Order Process)

Actor: Customer / System

Actor Actions	System Responses
The customer confirms the checkout	The system validates the shopping cart
The system verifies successful payment	The system creates a new order in the database
The system assigns an order ID	The system sets the initial order status to "Paid"
The system saves the order	The system displays an order confirmation page
The order is created successfully	The system sends a confirmation notification to the customer

Use Case: Order Ready (Update Order Status to Ready)

Actor: Employee / System

Actor Actions	System Responses
The employee finishes preparing an order	The system updates the order status to "Ready"
The system updates the status	The system saves the change in the database
The order status becomes "Ready"	The system sends a notification to the customer

Use Case: Mark Order as Delivered

Actor: Employee / Admin

Actor Actions	System Responses
The employee selects a delivered order	The system displays the order details
The employee clicks "Mark as Delivered"	The system updates the order status to "Delivered"
The status is updated	The system saves the change in the database

Use Case: Product Management

Actor: Admin

Actor Actions	System Responses
The admin opens product management	The system displays the product list
The admin adds or edits a product	The system saves the product changes
The admin deletes a product	The system removes the product from the system

Use Case: View Complaints

Actor: Admin

Actor Actions	System Responses
The admin opens the complaints page	The system displays all customer complaints
The admin selects a complaint	The system shows the complaint details
The admin marks complaint as handled	The system updates the complaint status

Use Case: Employee Management

Actor: Admin

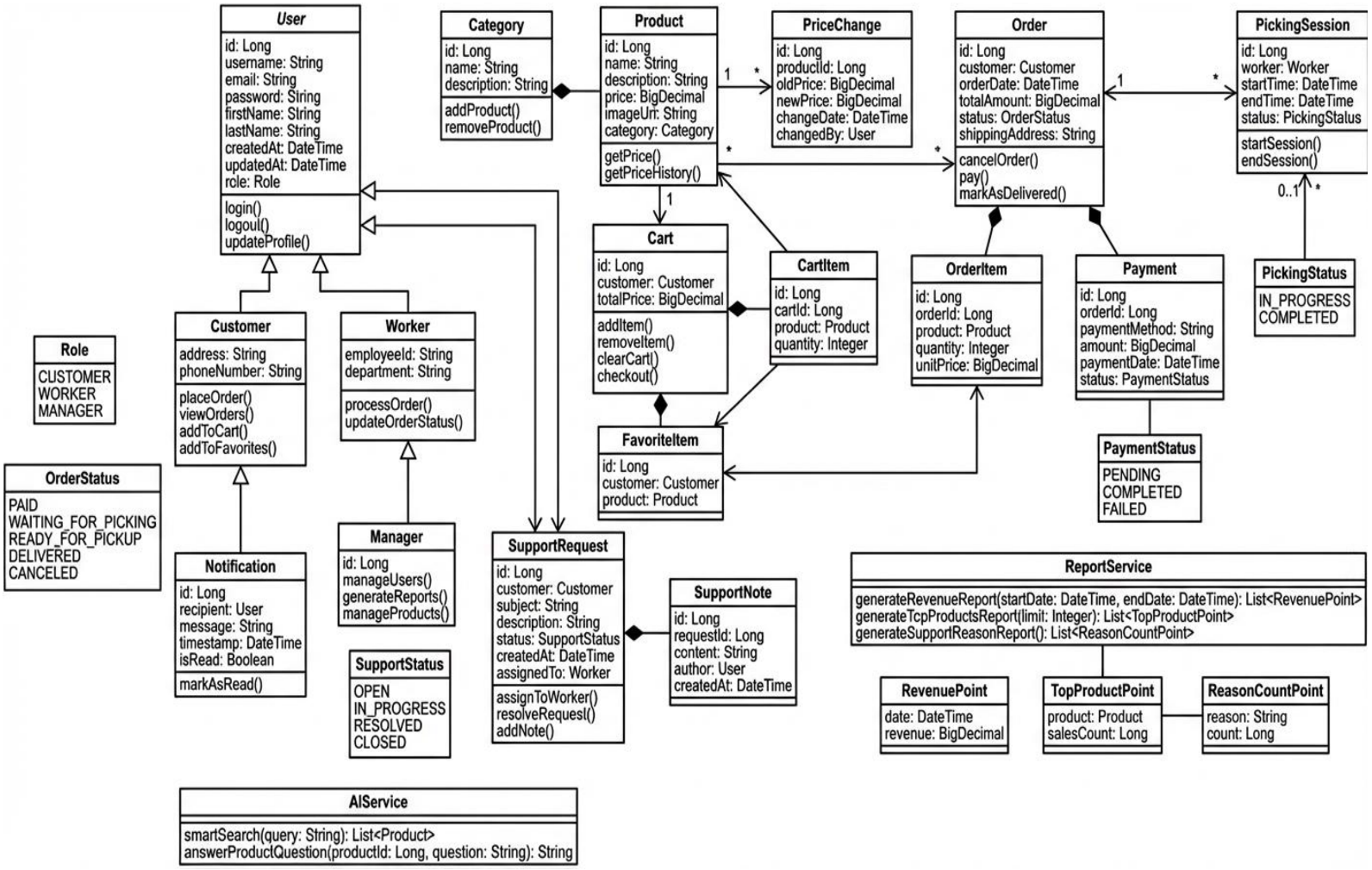
Actor Actions	System Responses
The admin opens the employee management page	The system displays the list of employees
The admin adds or edits an employee	The system saves the employee information
The admin deletes an employee	The system removes the employee from the system

Use Case: View Reports & Analytics

Actor: Admin

Actor Actions	System Responses
The admin opens the reports page	The system displays sales and system reports
The admin selects a report type	The system displays the selected analytics data
The admin filters the report by date or category	The system updates the displayed report

6.UML Class Diagram – Online Shopping System



This UML class diagram describes a complete e-commerce system. The core entity is User, which is extended by Customer, Worker, and Manager, each with different responsibilities (placing orders, processing orders, and managing the system and reports). Products are organized into Category objects, and each Product maintains a price and a history of price changes through PriceChange.

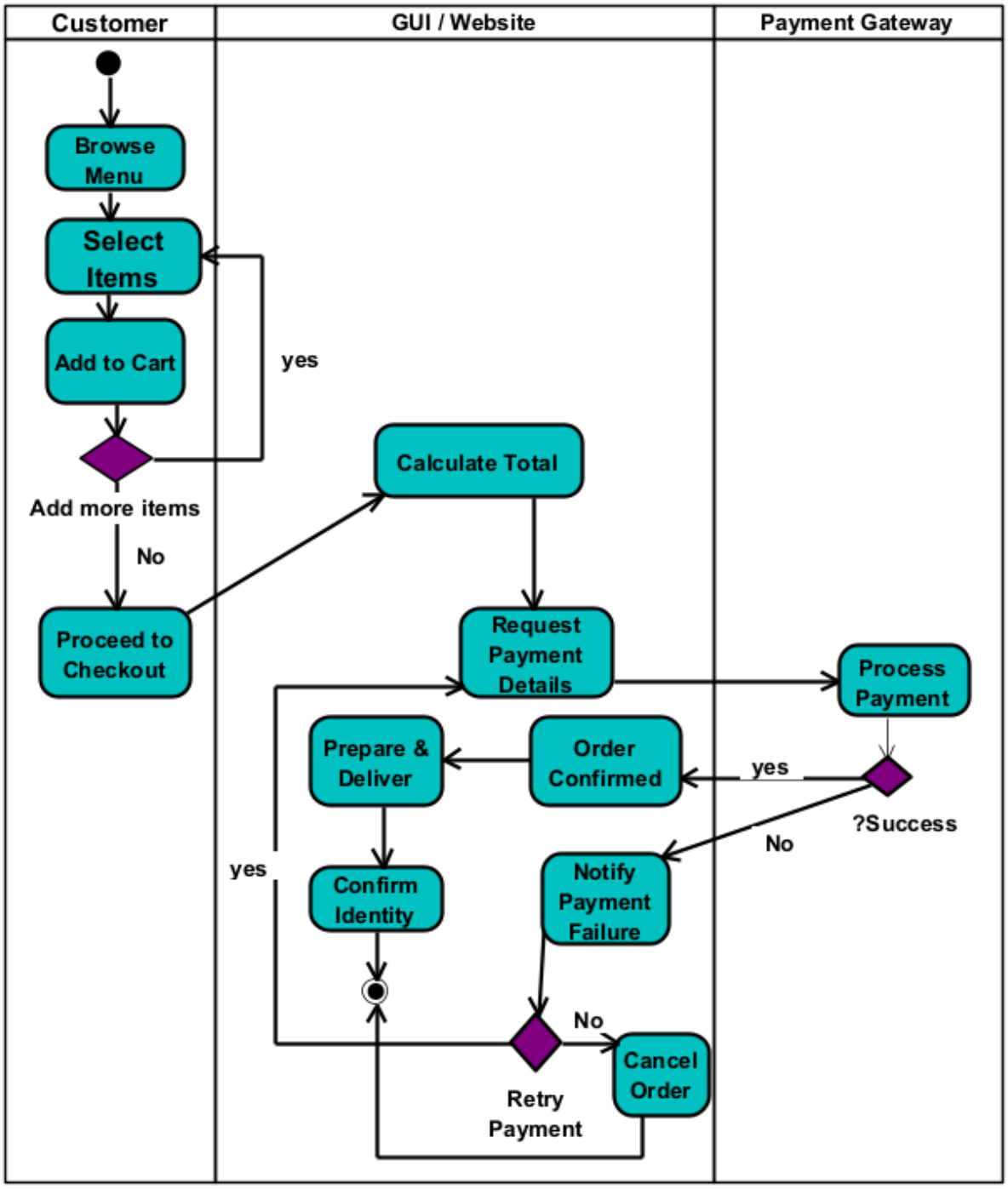
Customers can add products to a Cart, which contains multiple CartItem objects, and they can also save products as FavoriteItem. When checking out, a Cart is converted into an Order, which consists of multiple OrderItem entries and is associated with a Payment. Orders and payments are tracked using status enums (e.g., OrderStatus, PaymentStatus).

Warehouse processing is modeled using PickingSession, which represents the picking process performed by workers. The system also includes a customer support module with SupportRequest and SupportNote, allowing issues to be assigned, tracked, and resolved. Users can receive Notification messages about important events.

For management and analytics, the ReportService generates reports such as revenue over time, top-selling products, and support request reasons, using helper data classes (RevenuePoint, TopProductPoint, ReasonCountPoint). Finally, the AIService provides smart product search and answers product-related questions.

Overall, the diagram models the main business entities, their relationships, and the key processes of an online shopping platform: browsing products, purchasing, payment, logistics, support, and reporting.

7.1Order Process Activity Diagram Overview



This diagram illustrates the end-to-end workflow of a customer ordering process, integrated with a payment gateway. It is divided into three functional swimlanes: **Customer**, **GUI / Website**, and **Payment Gateway**.

1. Customer Lane (User Actions)

- **Initial Node:** The process begins with the customer browsing the menu.
- **Selection Loop:** The customer selects items and adds them to the cart. A **Decision Diamond** ("Add more items?") creates a loop:
    - **Yes:** Returns the user to select more items.
    - **No:** Proceeds to the checkout phase.
- **Checkout Initiation:** The user triggers the "Proceed to Checkout" action.

2. GUI / Website Lane (System Processing)

- **Calculation:** The system calculates the total amount including taxes and fees.
- **Data Entry:** The website requests payment details from the customer.
- **Order Confirmation:** Once payment is authorized, the system confirms the order.
- **Fulfillment:** The system initiates the "Prepare & Deliver" phase.
- **Final Verification:** The "Confirm Identity" step ensures the order reaches the correct recipient before reaching the **Final Node**.

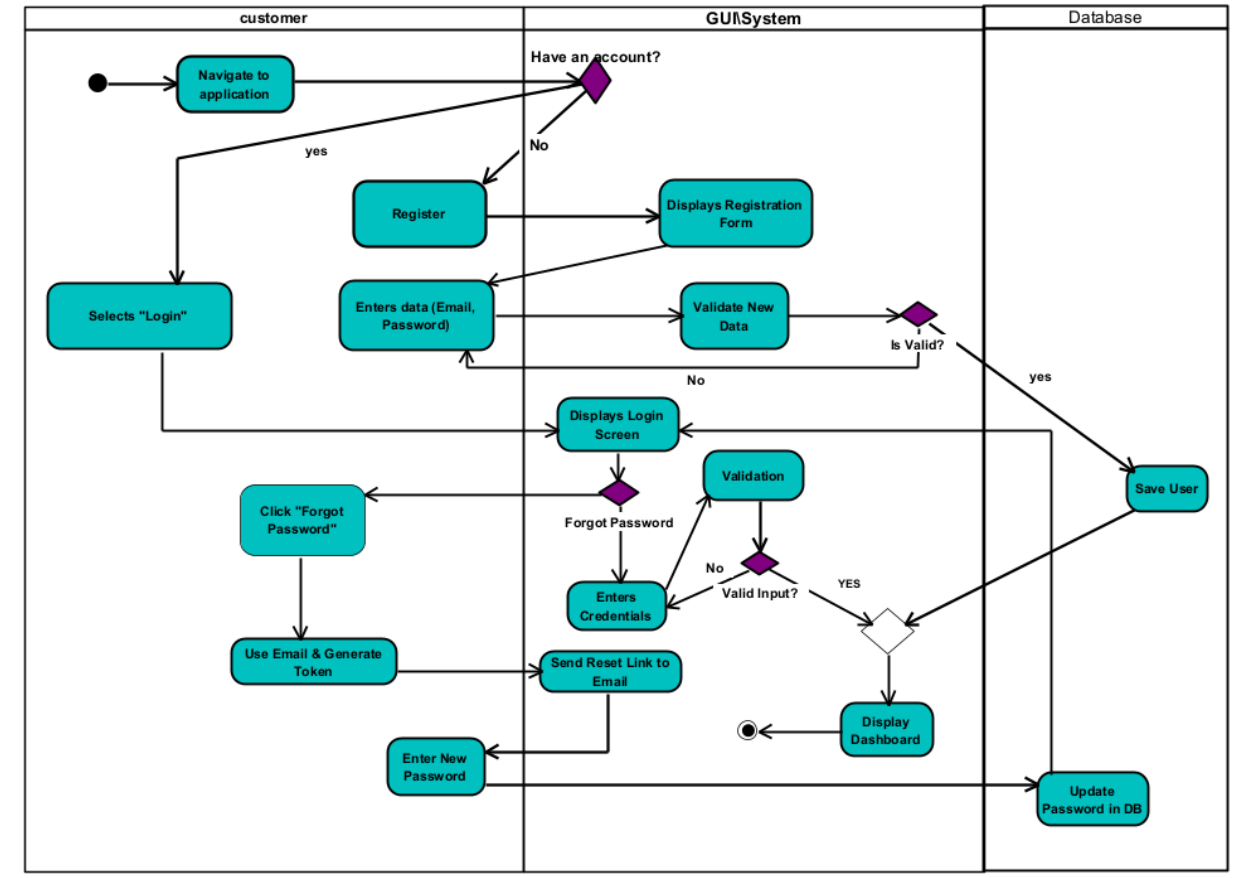
3. Payment Gateway Lane (External Verification)

- **Process Payment:** The system sends the encrypted data to the gateway to authorize the transaction.
- **Success Decision (?Success):**
    - **Yes:** The flow moves to "Order Confirmed."
    - **No:** The flow moves to "Notify Payment Failure."

4. Error Handling & Recovery (Exception Flow)

- If a payment fails, a **Retry Payment** decision point is triggered:
    - **Yes:** A loop redirects the user back to "Request Payment Details" to try a different card or update information.
    - **No:** The user chooses to terminate the process, leading to the **"Cancel Order"** action and the end of the activity.

7.2User Authentication and Registration Process



Technical Presentation: Customer Authentication & Management Flow

1. Architectural Overview (Swim lanes)

This Activity Diagram follows a multi-tier architecture, clearly separating responsibilities into three logical lanes:

- **Customer Lane**: Represents all user-facing interactions and inputs.
- **GUI/System Lane**: Manages the application logic, input validation, and user interface transitions.
- **Database Lane**: Handles the persistence layer, ensuring that all account changes (registrations and updates) are securely stored in the system's memory.

2. Logical Use Cases

The diagram effectively manages three critical workflows for the commercial website:

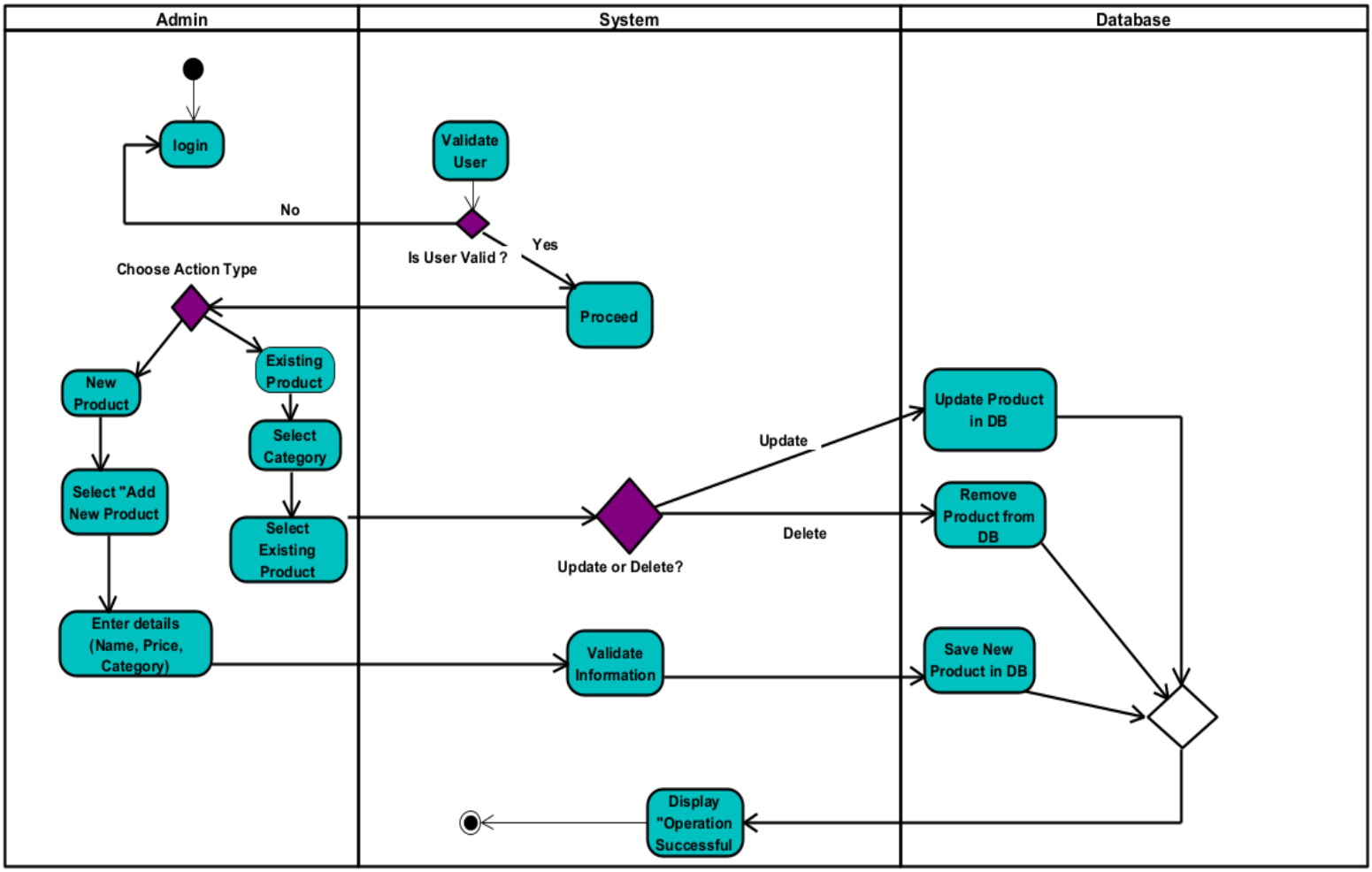
- **User Registration**: New customers can create an account. The system validates the input before the **Database** executes a Save User command.
- **Secure Authentication**: Existing users can log in. I have implemented a **Validation Loop** where the system checks the credentials; if they are invalid, a **"No" path** returns the user to the input stage to try again.
- **Password Recovery**: For users who forget their credentials, the flow triggers a secure reset process. Once the new password is provided, the **Database** performs an Update action, and the user is redirected back to the Login screen to ensure continuity.

3. UML Compliance & Advanced Logic

To ensure the diagram meets professional standards, I have incorporated specific UML structural elements:

- **Merge Node Implementation**: I replaced potential deadlocks with a **Merge Node** (the empty diamond) before the Display Dashboard state. This ensures that the flow proceeds correctly regardless of whether the user arrived via the Login or Registration path.
- **Database Persistence**: By including a dedicated Database lane, the diagram demonstrates that the system doesn't just "process" data but permanently stores it, which is vital for managing customer orders and profiles in a commercial environment.
- **State Termination**: Every path is accounted for, ensuring the process always reaches a clear **Final State** (the circled dot) after a successful operation.

7.3Admin Product Management Activity Diagram



This Activity Diagram models the backend management process for our commercial beauty and personal care website. It illustrates the workflow for an administrator performing CRUD (Create, Read, Update, Delete) operations on products like shampoos and perfumes.

1. Authentication & Guard Logic

- **Initial State**: The process begins with the Admin attempting to login.
- **System Validation**: The flow moves to the **System** lane, where the Validate User action occurs.
- **Decision Node**: A decision diamond evaluates the credentials. If the user is invalid, the flow loops back to the login state; otherwise, it proceeds to the dashboard.

2. Admin Decision Making (Action Selection)

- **Action Type**: The Admin reaches a decision point to either handle a New Product or work with an Existing Product.
- **Product Selection**:
    - For existing items, the Admin filters by Select Category before choosing a specific product.
    - For new items, the Admin enters essential metadata: **Name, Price, and Category**.

3. Data Processing & Persistence

- **Input Validation**: Before any database commit, the **System** performs Validate Information to ensure data integrity.
- **Database Lane**: This lane handles the final persistence layer. The system branches into three distinct database operations: Update, Remove, or Save New Product.

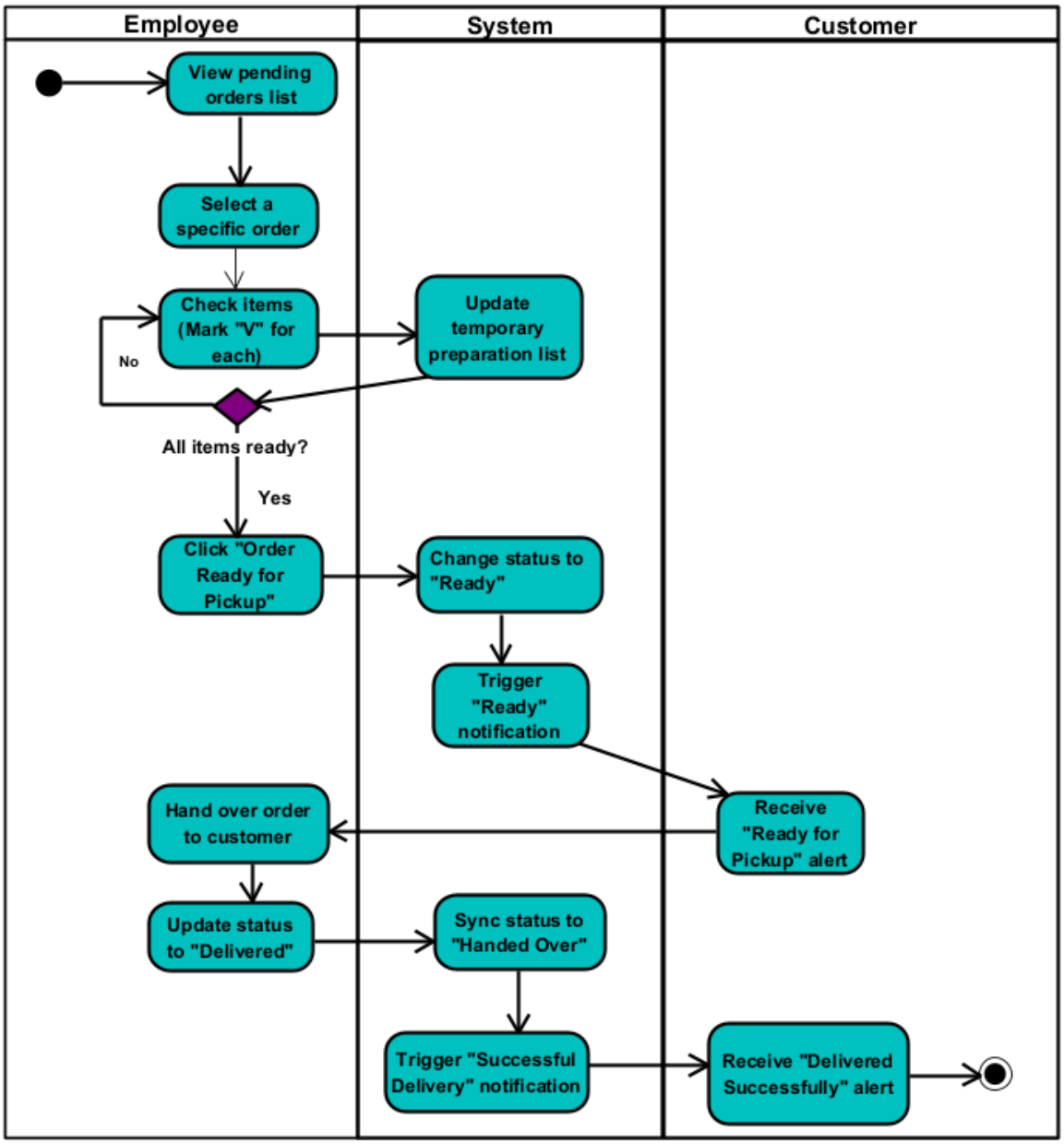
4. Structural Correctness: The Merge Node

- **Merge vs. Join**: I have specifically utilized a **Merge Node** (the diamond symbol) at the end of the Database lane.
- **Logical Flow**: In previous iterations, a "Join Bar" would have created a deadlock because it requires all three inputs to trigger. Since the Admin performs only **one** action per session (OR logic), the Merge Node correctly allows any single completed database operation to trigger the final success message.

5. Conclusion & Feedback

- **Final Feedback**: Once the database confirms the action, the **System** lane triggers the Display "Operation Successful" notification.
- **Final State**: The process terminates successfully.

7.4Technical Presentation: Order Fulfillment & Handover Workflow



1. Operational Architecture (Swim lanes)

This diagram illustrates the internal logistics of the commercial platform, utilizing three distinct lanes to define role-based responsibilities:

- **Employee**: Represents the warehouse or store staff managing the physical preparation and handover of products.
- **System**: Acts as the central backend, managing order statuses and automated communications.
- **Customer**: Represents the end-user receiving real-time updates and the physical order.

2. Logical Phases & Process Integrity

The workflow is divided into two main synchronization phases:

- **Preparation Phase**: The employee performs an inventory check. I have included a **Decision Node** ("All items ready?") to ensure that the order status cannot be changed to "Ready" until every item is physically verified.
- **Handover Phase**: Once the customer arrives, the employee updates the status to "Delivered." The **System** then synchronizes this with the "Handed Over" status, providing a digital audit trail for the transaction.

3. Automated Notification System

A key feature of this design is the integration of system-triggered alerts:

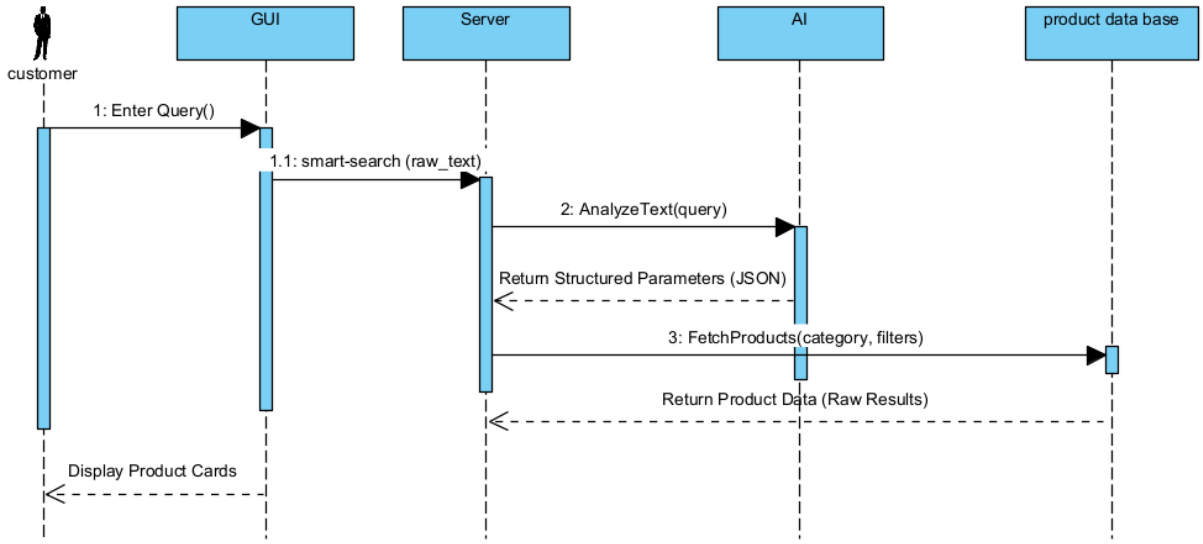
- When the employee clicks "Order Ready for Pickup," the **System** automatically notifies the **Customer**.
- After the final handover, a "Successful Delivery" notification is triggered.
- This automation ensures a seamless user experience, which is a core requirement for our commercial beauty and personal care website.

4. Technical Correctness

The diagram is now technically robust:

- **Elimination of Redundancy**: Previous duplicate alert boxes in the Employee lane have been replaced with the functional "Hand over order to customer" and "Update status" actions.
- **Process Termination**: The flow reaches a definitive **Final State** only after the system confirms the successful delivery to the customer.

8.1Technical Specification: AI-Powered Smart Search Sequence



1. Executive Summary

This Sequence Diagram illustrates one of the core innovative features of the commercial platform: the **Smart Search** engine. Unlike traditional search functions that rely on simple keyword matching, this system leverages **Artificial Intelligence (AI)** to understand user intent and provide highly relevant results for beauty and personal care products.

2. Object Lifeline Definitions

The diagram consists of five key participants in the search lifecycle:

- **Customer**: The end-user who initiates the interaction by entering a natural language query.
- **GUI (Graphical User Interface)**: The front-end layer that captures user input and displays the final product results.
- **Server**: The central orchestrator that manages communication between the user interface, the AI engine, and the database.
- **AI (Artificial Intelligence Engine)**: The intelligence layer responsible for Natural Language Processing (NLP) and intent analysis.
- **Product Database**: The persistence layer where all product information (shampoo, soaps, perfumes, etc.) is stored.

3. Detailed Process Flow

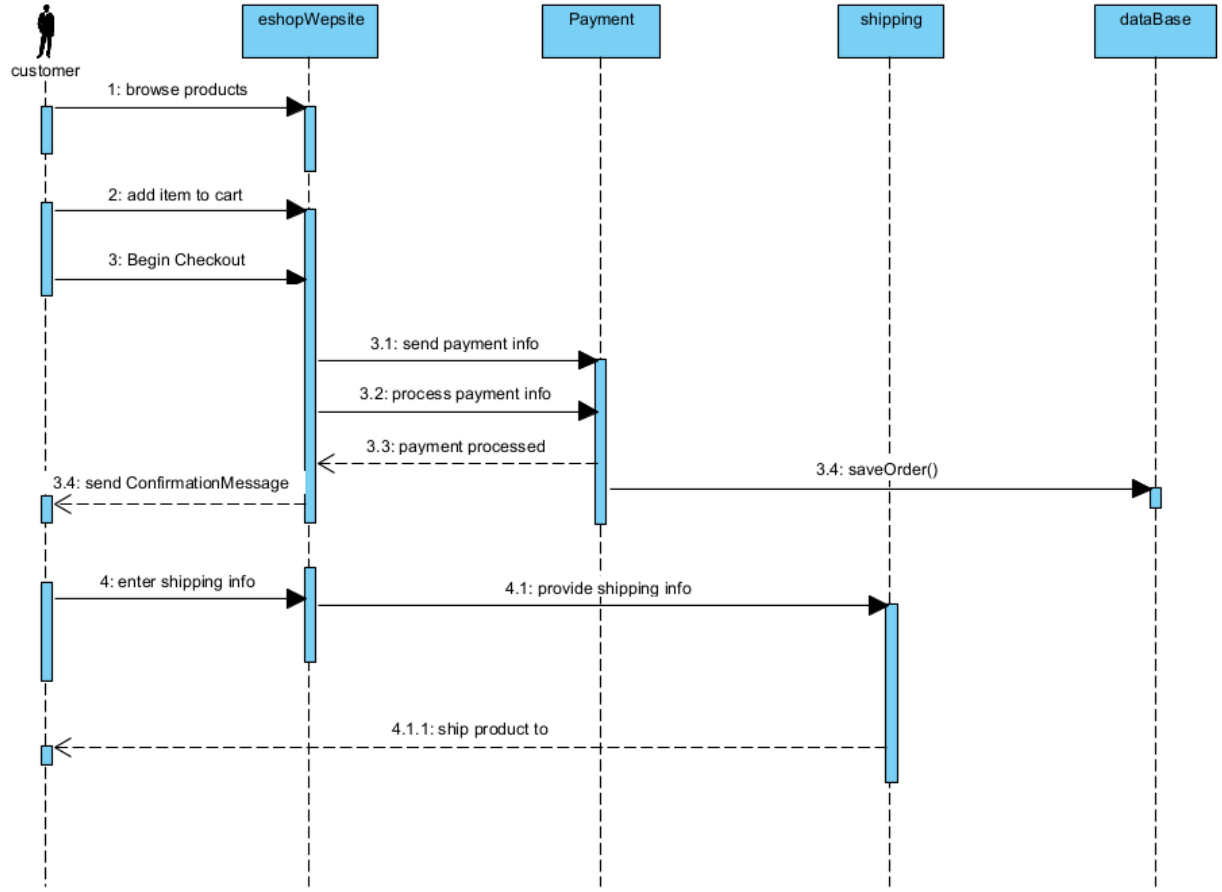
1. **Query Initiation**: The **Customer** submits a query through the Enter Query() method on the **GUI**.

2. **Data Transmission**: The **GUI** forwards the raw text to the **Server** via the smart-search(raw_text) call.
3. **Intent Analysis**: The **Server** sends the text to the **AI** using the AnalyzeText(query) message. The AI breaks down the sentence to identify specific product attributes (e.g., identifying "dandruff" as a medical filter and "shampoo" as the category).
4. **Structured Data Mapping**: The **AI** returns the analyzed data as a **JSON (Structured Parameters)** object back to the **Server**, transforming human language into machine-readable filters.
5. **Optimized Database Query**: Using the AI-generated parameters, the **Server** executes a FetchProducts(category, filters) call to the **Product Database**.
6. **Result Rendering**: The **Database** returns the relevant product data, and the **GUI** displays "Product Cards" to the user, providing a personalized and accurate shopping experience.

4. Technical Design Principles

- **Separation of Concerns**: By placing the AI as a separate lifeline, the system remains modular and scalable.
- **UML Compliance**: The diagram utilizes solid arrows for synchronous requests and dashed arrows for return messages to ensure technical accuracy.
- **Commercial Value**: This flow directly supports the sales of specific products like perfumes and shampoos by ensuring the customer finds the most relevant item quickly, increasing conversion rates.

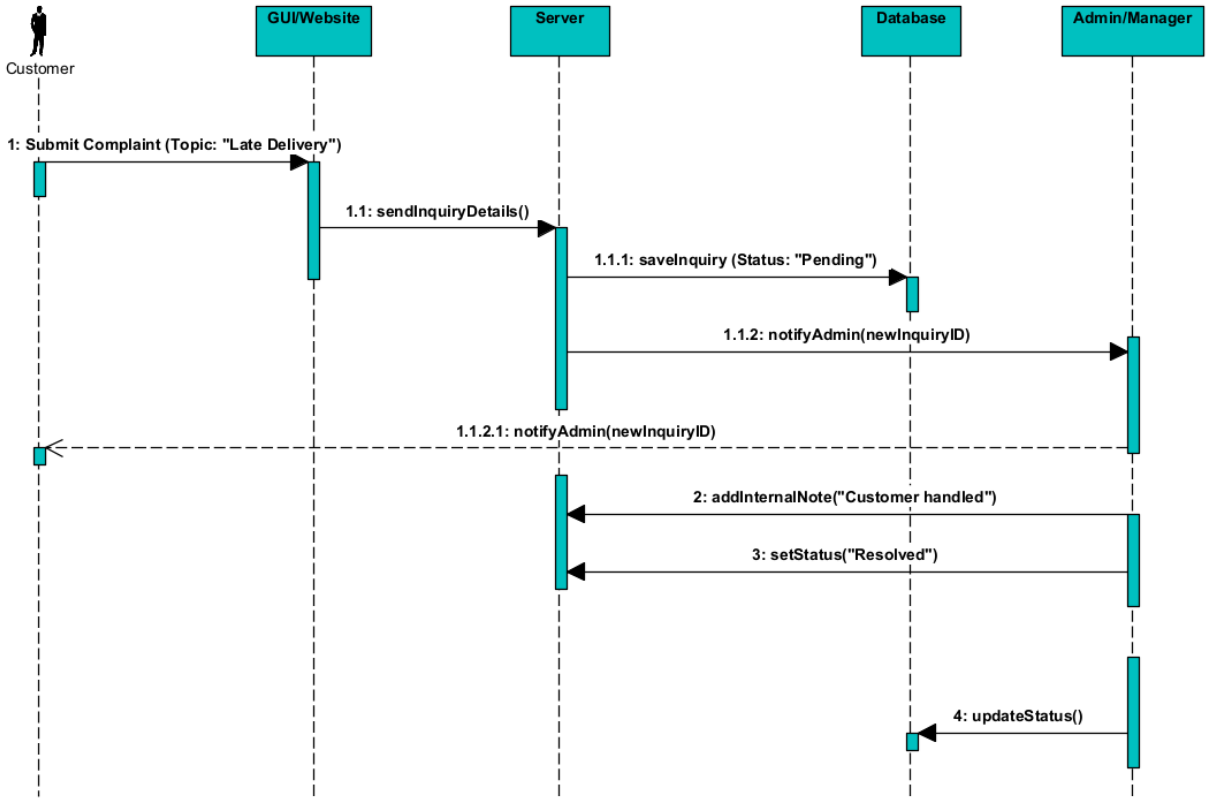
8.2Technical Specification: Order & Checkout Sequence Diagram



"This Sequence Diagram illustrates the complete purchase lifecycle of the commercial platform, ensuring data integrity and secure transaction handling.

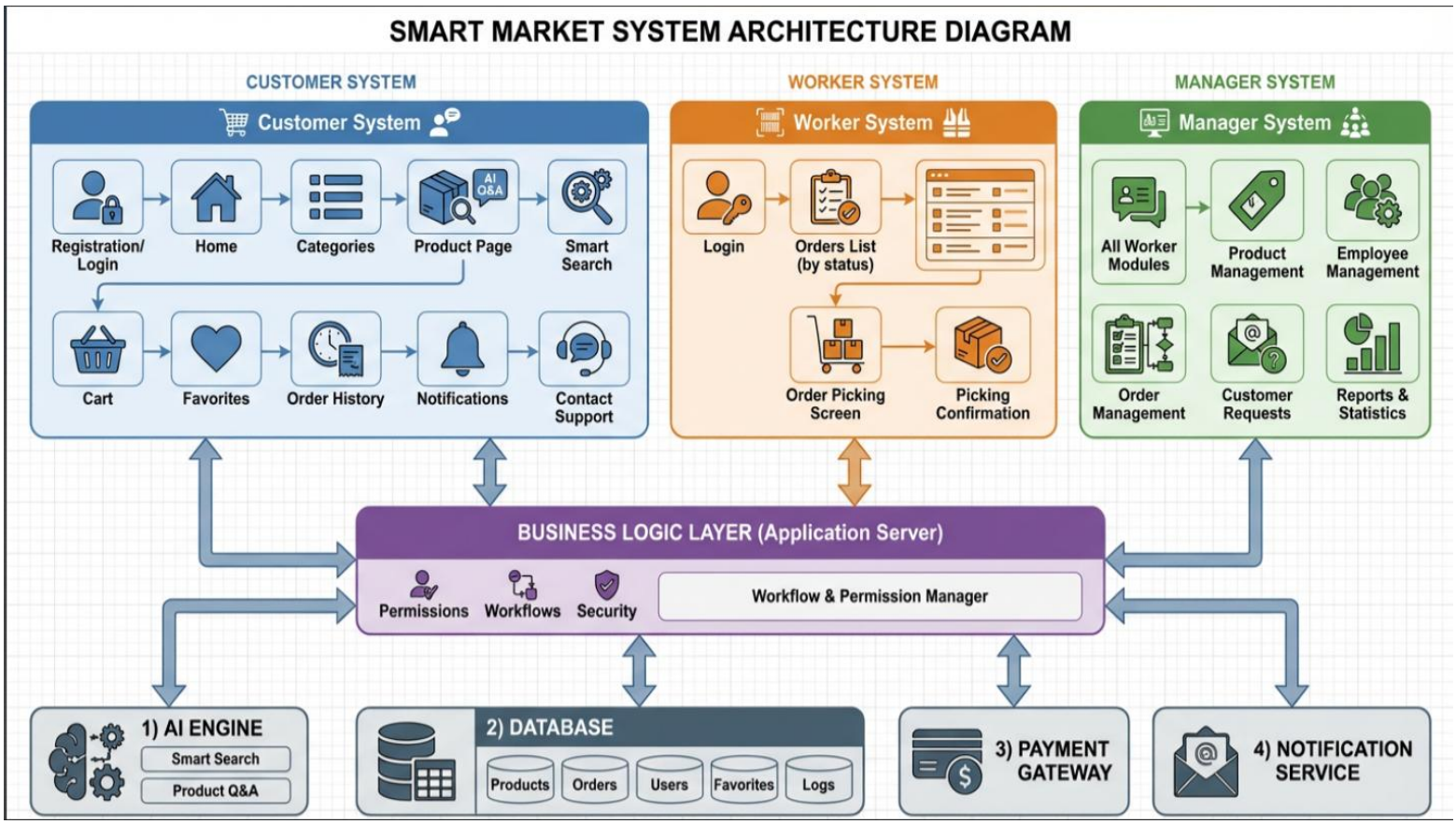
- **Transaction Validation**: The system is designed to wait for a synchronous 'payment processed' response from the external **Payment Gateway** before proceeding. This prevents order confirmation without successful billing.
- **Data Persistence**: A dedicated **dataBase** lifeline has been integrated. The saveOrder() method ensures that once payment is verified, the order is permanently recorded. This allows the internal warehouse staff to access and fulfill the order later.
- **Separation of Concerns**: The diagram clearly separates front-end interactions (GUI), backend orchestration (eshopWebsite), and external service providers (Payment and Shipping), following standard enterprise architecture principles.
- **Automated Feedback**: The workflow concludes by providing the **Customer** with both a digital confirmation and a shipping update, ensuring a seamless user experience for purchasing beauty and personal care products."

8.3Technical Specification: Manual Support Handling Workflow



- **Inquiry Persistence**: The system ensures all customer inquiries are immediately logged in the **Database** with a "Pending" status, preventing data loss.
- **Administrative Oversight**: The **Admin Dashboard** notifies the manager, who then initiates external contact (phone/email) with the customer to resolve the issue.
- **Internal Documentation**: A key feature of this workflow is the addInternalNote function, allowing the manager to document the resolution details internally before closing the case.
- **Final Resolution**: The process concludes with a manual status update to "Resolved." This updates the **Database** records, marking the commercial task as completed within the system's backend.

9.System Architecture



Smart Market System Architecture Explanation

The Smart Market system is an e-commerce platform that supports three types of users: Customer, Worker, and Manager. The system is designed using a layered architecture, which provides a clear separation between the presentation layer, the business logic layer, and the infrastructure/services layer. This design improves security, maintainability, and supports future scalability of the system.

1. Presentation Layer

This layer includes three main subsystems:

Customer System – This subsystem allows customers to register and log in, browse categories and products, use smart search, manage their cart and favorites, view their order history, receive notifications, and contact support. It is responsible for the customer's user experience and for presenting data in a clear and user-friendly way.

Worker System – This subsystem is used by store employees. It allows them to log in, view order lists by status, perform order picking, and confirm the preparation of orders. This system

supports the operational workflow of the store and helps employees process customer orders efficiently.

Manager System – This subsystem is used by administrators and managers. It includes product management, employee management, order management, handling customer requests, and generating reports and statistics. This system provides full control and monitoring over the entire platform.

All these subsystems do not access the database directly. Instead, they communicate only with the application server.

2. Business Logic Layer (Application Server)

This is the core layer of the system, responsible for implementing the business rules and workflows. All requests coming from the presentation layer (Customer, Worker, and Manager systems) pass through this layer.

The responsibilities of this layer include:

- Managing permissions and role-based access control
- Managing business workflows
- Enforcing system security
- Coordinating communication between the presentation layer and the infrastructure services

This layer ensures that every action is executed according to the system's rules and the user's authorization level, which guarantees consistency, security, and correctness of the business processes.

3. Infrastructure and Services Layer

This layer contains the external services and core infrastructure components used by the system:

Database – Stores all the system data, including users, products, orders, favorites, and logs.

Payment Gateway – A secure external service responsible for processing customer payments.

Notification Service – A service that sends notifications to users, such as order status updates, "order ready for pickup" messages, and other system alerts.

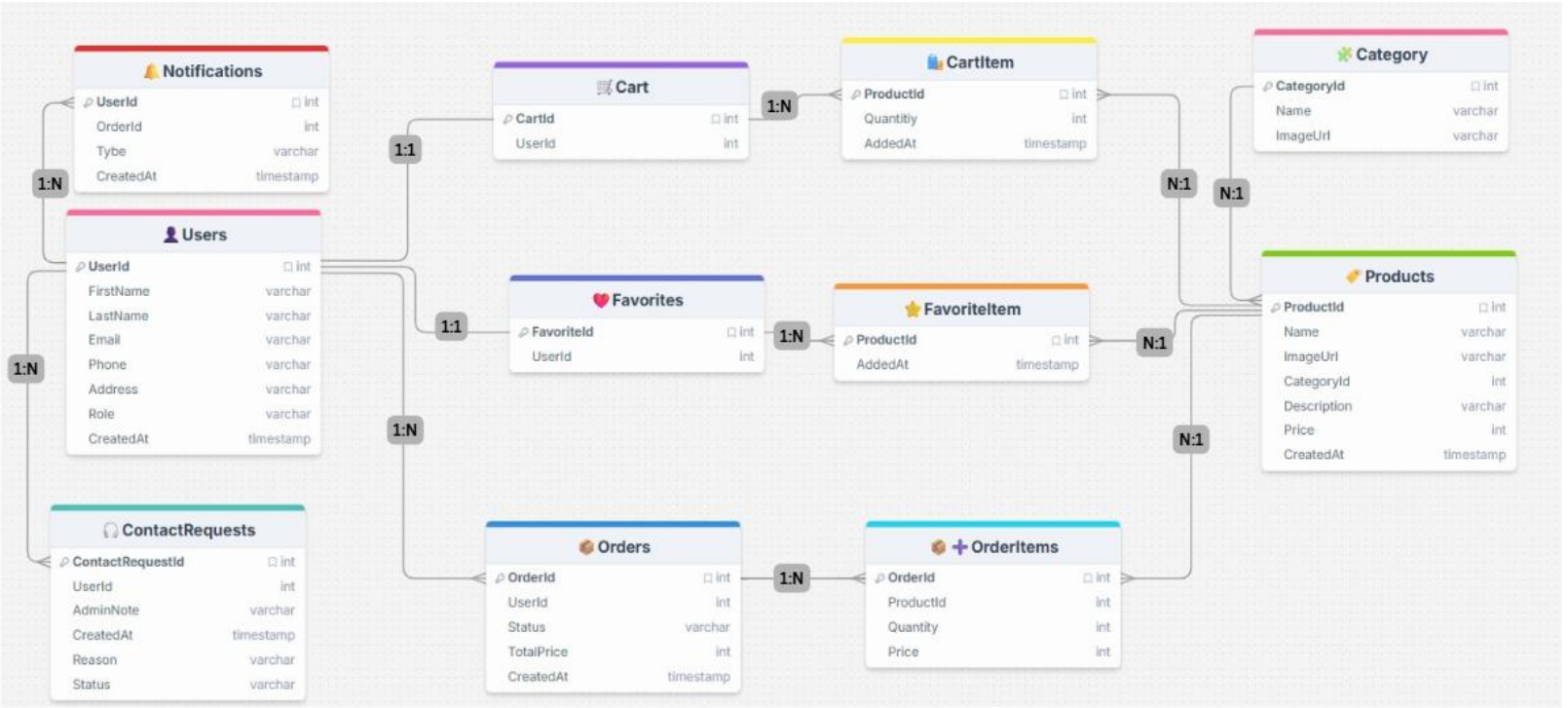
AI Engine – A smart engine that provides advanced features such as smart search and product Q&A, in order to improve the user experience.

4. Design Principles

The system architecture is based on the principles of separation of concerns, security, and scalability. This separation allows:

- Preventing direct access to the database from user interfaces, thus improving data security
- Easier maintenance and future extension of the system
- Independent scaling and development of each layer according to system need

10.SMART MARKET – System Database Design



Main Entities:

Users:

Stores all users in the system (customers, workers, admins) with their personal data and role.

Relationships:

- Each user has **one Cart** (1:1)
- Each user has **one Favorites list** (1:1)
- A user can have **many Orders** (1:N)
- A user can send **many ContactRequests** (1:N)
- A user can receive **many Notifications** (1:N)

Category:

Stores product categories.

Relationship:

- One category can contain **many Products** (1:N)

Products:

Stores all products with name, description, price, and image.

Relationships:

- Each product belongs to **one Category** (N:1)
- A product can appear in many:
    - CartItems
    - FavoriteItems
    - OrderItems

Cart and Favorites:

Cart:

Each user has exactly one cart.

Relationship:

- One Cart contains **many CartItems** (1:N)

CartItem:

Represents one product inside a cart, with quantity and added date.

Relationships:

- Each CartItem belongs to **one Cart**
- Each CartItem references **one Product** (N:1)

Favorites:

Each user has one favorites list.

Relationship:

- One Favorites list contains **many FavoriteItems** (1:N)

FavoriteItem:

Represents one product saved in the favorites list.

Relationship:

- Each FavoriteItem references **one Product** (N:1)

Orders:

Orders:

Stores orders made by users, including status, date, and total price.

Relationships:

- User 1:N Orders (one user can make many orders)
- Order 1:N OrderItems (one order contains many items)

OrderItems:

Represents one product inside an order, with quantity and price at purchase time.

Relationship:

- Each OrderItem references **one Product** (N:1)

Admin System:

ContactRequests:

Stores support/contact requests sent by users.

Relationship:

- User 1:N ContactRequests

Notifications:

Stores notifications sent to users (e.g., “order ready for pickup”).

Relationships:

- User 1:N Notifications

11.Black Box Test Plan – SMART MARKET

Test ID	Test Description	Input	Expected Result	Actual Result	status
1	Login with valid credentials	Valid email and password	User is logged into the system	Logged in successfully	**passed**
2	Login with invalid password	Valid email, wrong password	Error message is displayed	Error message displayed	**passed**
3	User registration	Valid user details (first name, last name, email, password, etc.)	User is created and all details are saved in the database	User created and all details saved	**passed**
4	Product search (free text)	"anti dandruff shampoo"	Relevant products are displayed	Relevant products displayed	**Passed**
5	Open category	Click on a category	Category products are displayed	Products displayed	**Passed**
6	Validate products have required fields (name, price, image)	Load product list	All products have name, price and image	All products contain required fields	**Passed**
7	Open product page	Click on a product	Product details page is opened	Product page opened	**Passed**
8	Add product to cart	Click "Add to Cart"	Product is added to cart	Product added to cart	**Passed**
9	Add product to favorites	Click "Add to Favorites"	Product is saved in favorites	Product saved in favorites	**Passed**
10	Create order	Click "Place Order"	Order is saved in the system	Order saved successfully	**Passed**
11	View order from notification	Click on “View Order” from notification	Only the specific order related to the notification is displayed	Correct order displayed	**Passed**
12	View my orders	Open "My Orders"	User orders are displayed	Orders displayed	**Passed**

13	Send customer ticket	Select ticket type and submit	Ticket is saved in the system	Ticket saved	**Passed**
14	Worker views new order	Login as worker	New order appears in the list	Order appears in the list	**Passed**
15	Worker updates order status	Change order status	Status is updated in the system	Status updated	**Passed**
16	Admin adds new product	Fill new product form	Product is added to the system	Product added successfully	**Passed**
17	Admin updates product price	Change product price	Price is updated in the system	Price updated	**Passed**
18	Admin deletes product	Click "Delete" on product	Product is removed from the system	Product deleted successfully	**Passed**
19	Redirect regular worker to picking screen	Login as regular worker	The worker is not exposed to the admin main page and is redirected directly to the picking system	The worker was redirected directly to the picking system	**Passed**
20	Admin access to picking system	Login as admin	Picking screen accessible	Screen opened	**Passed**
21	Admin access to product management	Login as admin	Product management accessible	Screen opened	**Passed**
22	Admin views analytics and graphs	Login as admin and open graphs page	Analytics dashboard is displayed	Graphs displayed successfully	**Passed**
23	Worker direct access to admin URL blocked	Worker manually enters admin URL in browser	Access denied / redirect	Access blocked successfully	**Passed**
24	Admin edits product details	Edit product details and save	Updated product details are saved in Firebase and reflected on website	Product updated successfully	**Passed**
25	Product appears on customer website	Product appears on customer website	Add new product as admin	Product visible on website	**Passed**

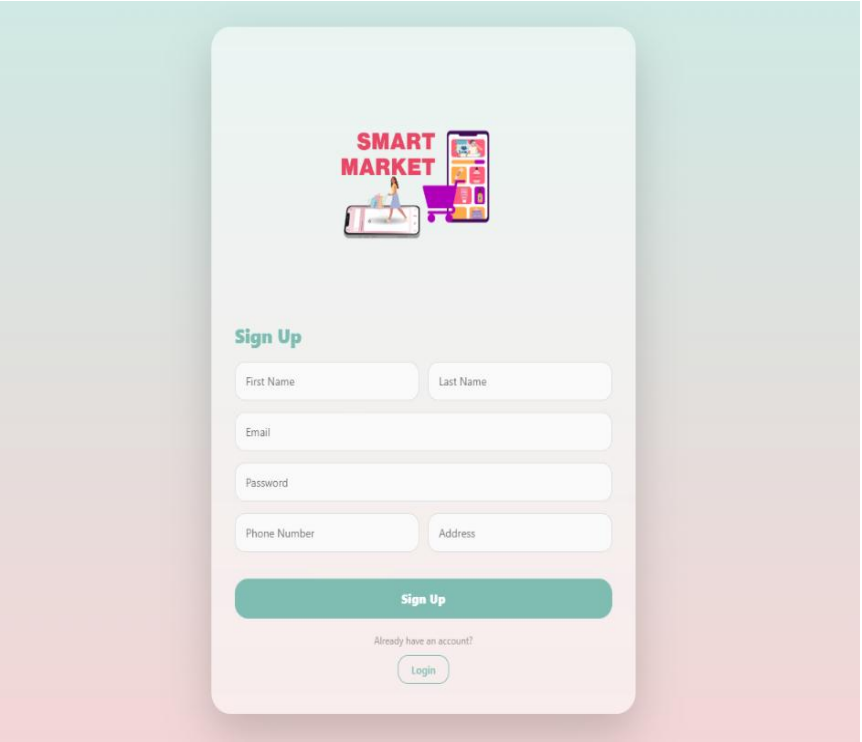
26	Order status sync between worker and customer	Worker changes order status	Customer sees updated order status	Status updated and reflected correctly	**Passed**
27	Product deletion reflected on website	Admin deletes product	Product is removed from customer website after deletion	Product removed from website	**Passed**
28	Order saved correctly in Firebase	Customer clicks "Place Order"	Order document created in Firebase with correct data	Order stored correctly in Firebase	**Passed**
29	Ticket saved correctly in Firebase	Customer submits contact request	Ticket document created in contact_requests collection	Ticket stored correctly in Firebase	**Passed**
30	Notification created after status change	Worker updates order status	Notification document created in notifications collection	Notification created in Firebase	**Passed**
31	Worker cannot access analytics	Worker tries to open analytics page URL	Worker cannot access analytics page	Access denied	**Passed**
32	Worker cannot access product management	Worker tries to open product management URL	Worker cannot access product management page	Access denied	**Passed**
33	Worker admin action restricted	Worker attempts to edit or delete product	Worker cannot perform admin product actions	Unauthorized action blocked	**Passed**

12.Application Screens – Overview and Explanations

Sign Up (Customer Registration)

Purpose:
The Sign Up screen allows new customers to create an account in the Smart Market system.

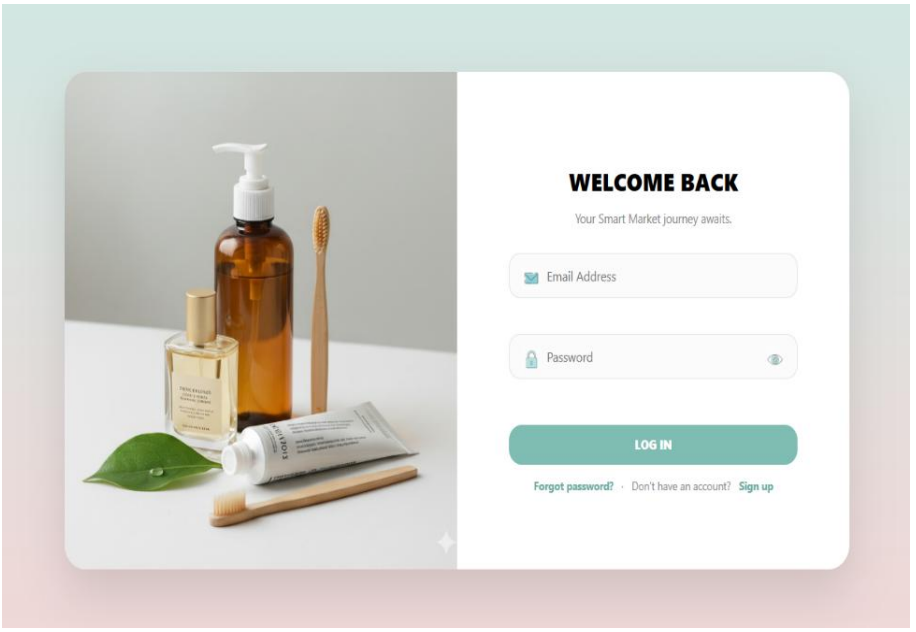
Description:
On this screen, the user enters personal details including first name, last name, email address, password, phone number, and address. After filling in the required fields, the user clicks the "Sign Up" button to create a new account. If the user already has an account, they can navigate to the Login screen using the "Login" button.



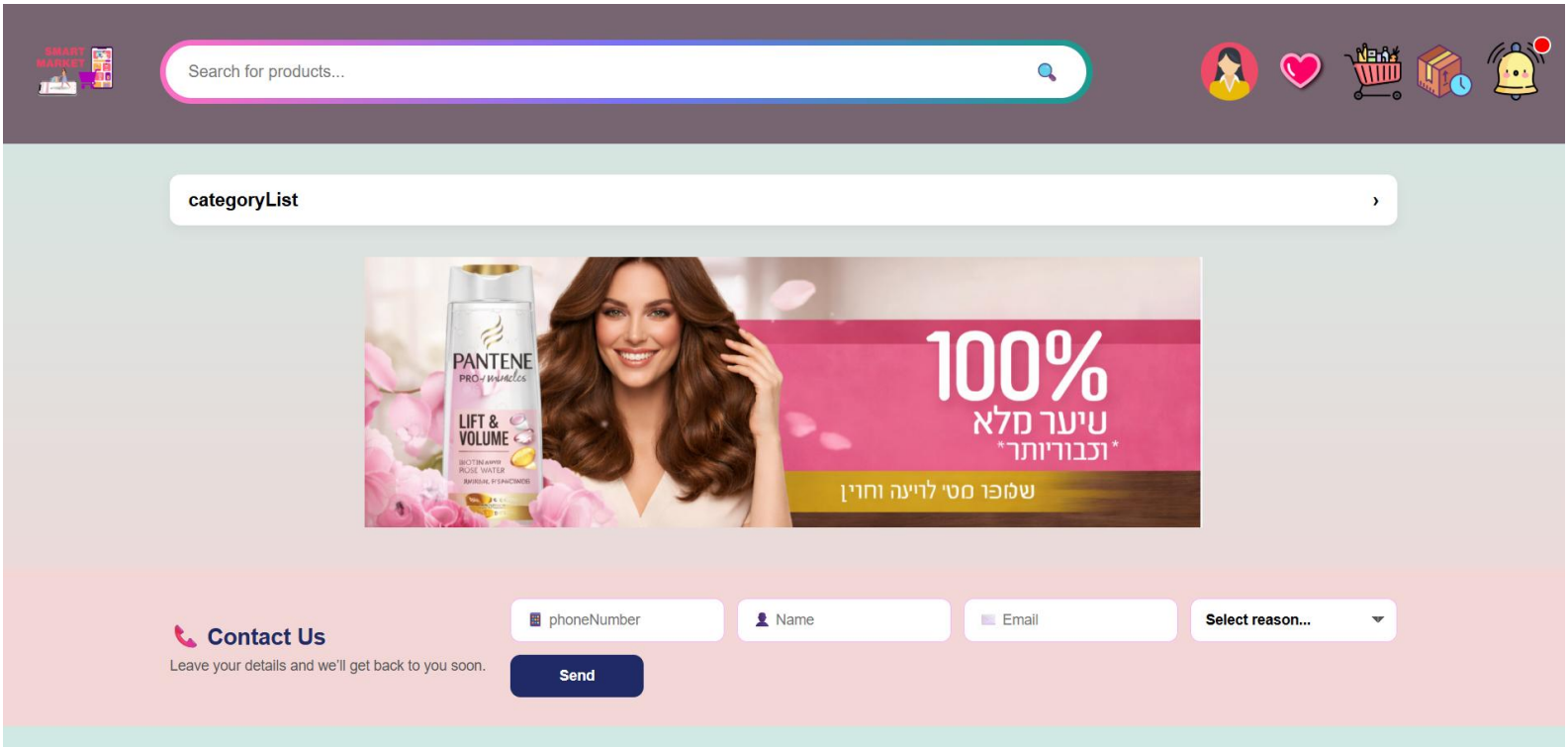
Login

Purpose:
The Login screen allows registered customers to access their accounts in the Smart Market system.

Description:
On this screen, the user enters their email address and password. After clicking the "Log In" button, the system verifies the credentials and, if they are correct, grants access to the system. The screen also provides options to recover a forgotten password and to navigate to the Sign Up screen for users who do not yet have an account.



13.Screen: Home / AI Product Search



Purpose:

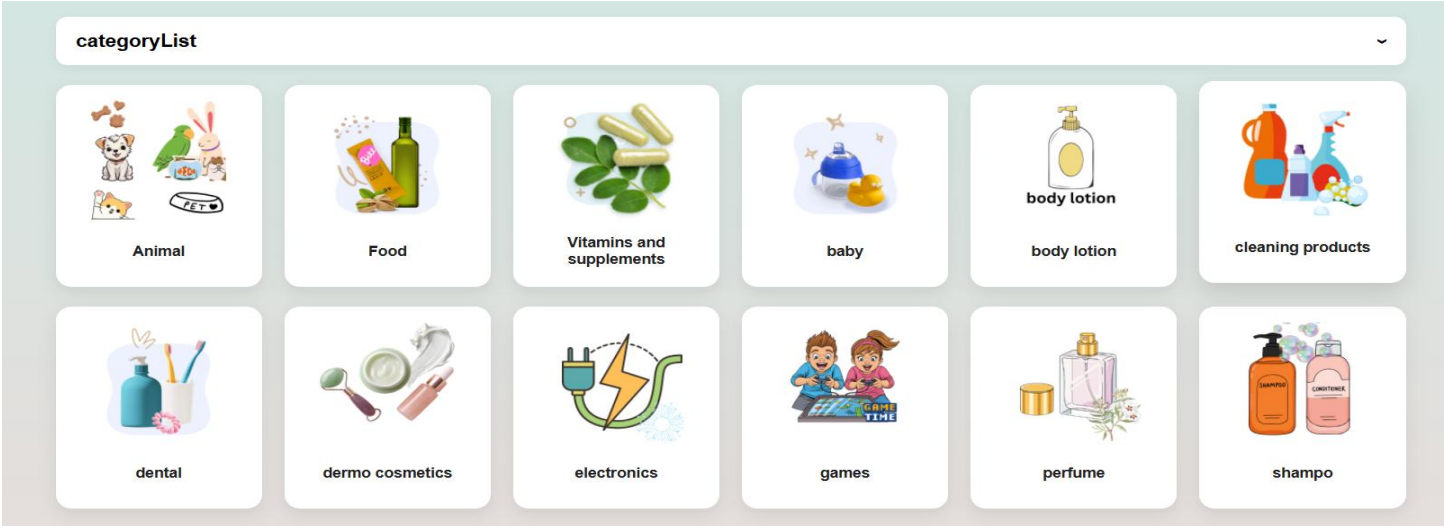
The Home screen allows customers to search for products using an AI-based intelligent search system and to browse products in the Smart Market platform.

Description:

This screen provides an advanced AI-powered search bar that enables users to search for products using natural language, similar to a chatbot experience. The system can understand user intent, product-related questions, and general queries, and then suggest relevant products accordingly.

Users can also browse products by categories using the category list. The top navigation bar provides quick access to the user profile, favorites, shopping cart, orders, and notifications.

In addition, the screen displays promotional banners and includes a "Contact Us" section that allows customers to send inquiries or complaints by entering their contact details and selecting a reason.

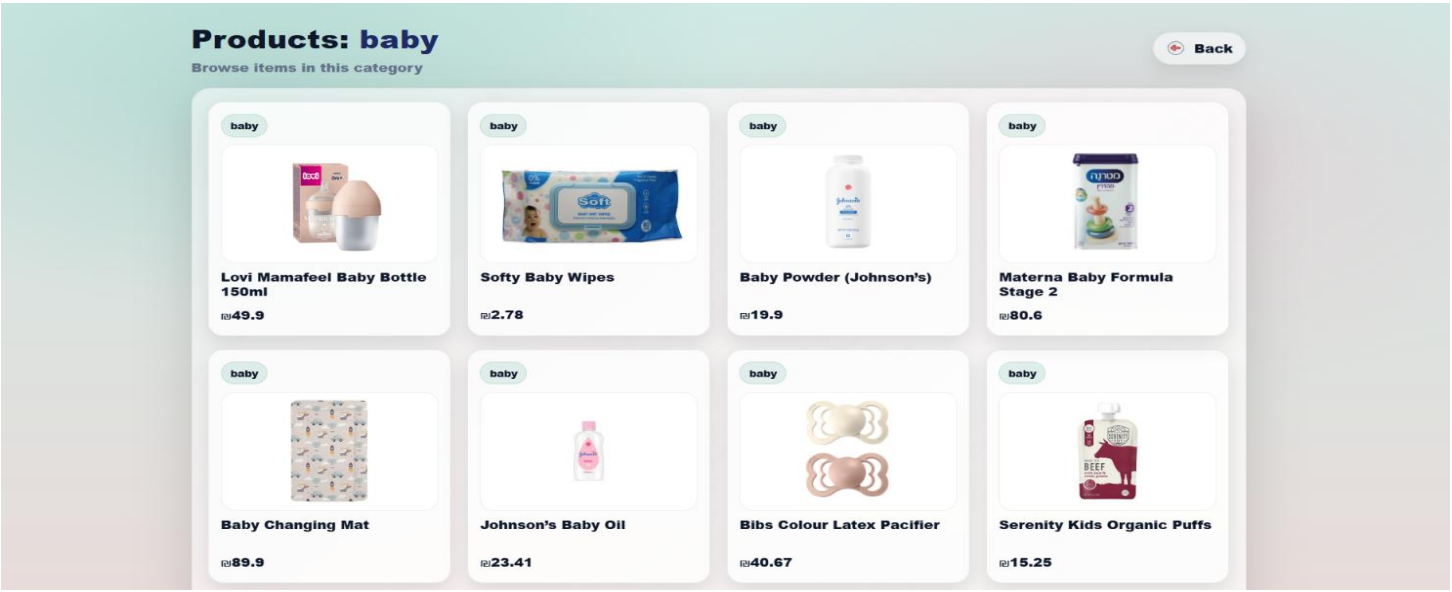


Description:

This screen displays a list of product categories such as Food, Baby, Electronics, Cosmetics, and others.

Each category is presented as a clickable card with an icon and a title. When the user selects a category, the system navigates to that specific category page.

The purpose of this screen is to help the customer easily understand how the store is organized and quickly choose the type of products they are interested in.



Category Products

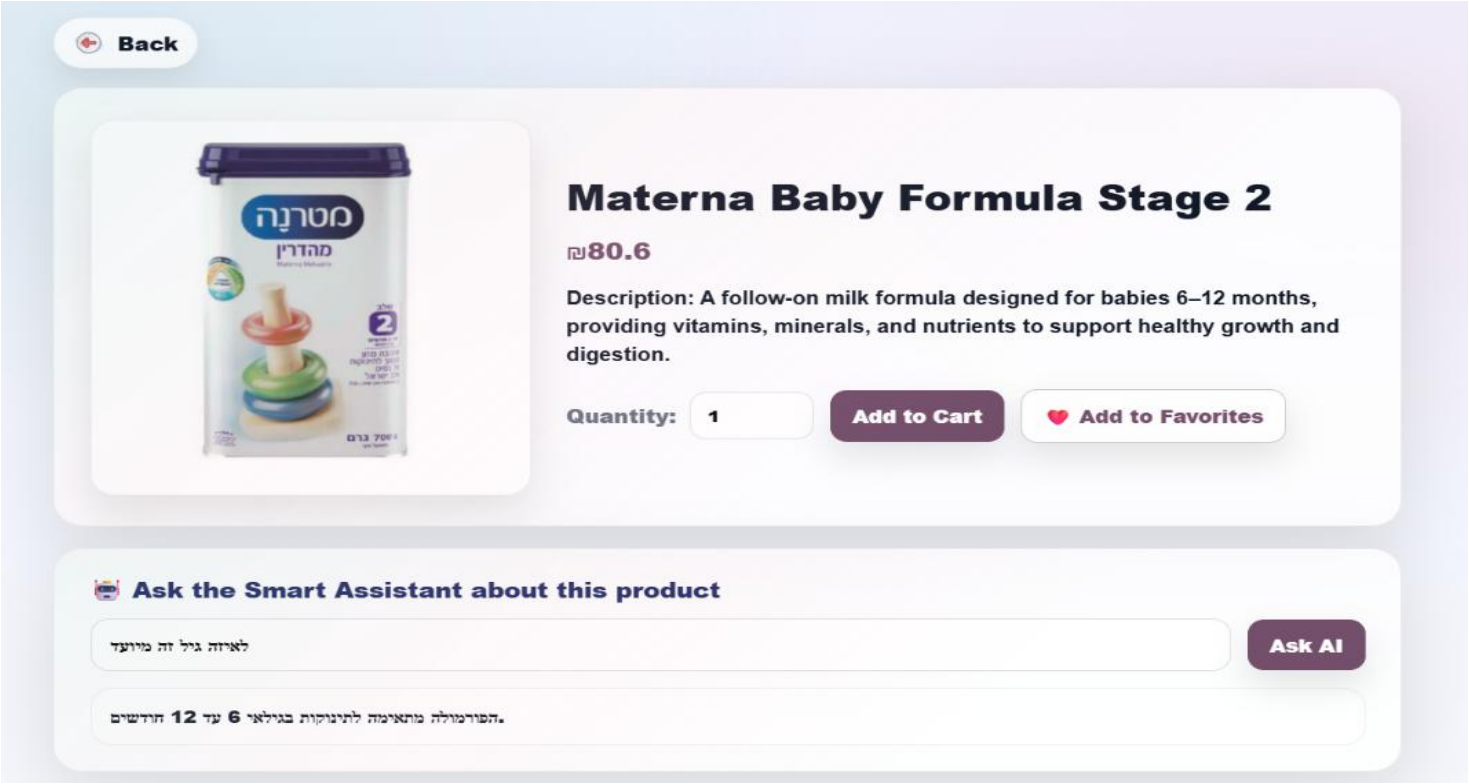
Description:

This screen displays a list of products from a specific category (for example, the "Baby" category).

Each product is presented as a card that includes an image, the product name, and the price.

The user can scroll through the list and browse the available items in this category.

A "Back" button is provided to allow the user to return to the categories screen.



Products page

Description:

This screen displays detailed information about a specific product, including the product image, name, price, and description.

The customer can choose the desired quantity and add the product to the shopping cart or to the favorites list.

In addition, the screen includes a Smart Assistant (AI) feature that allows the customer to ask questions about the product and receive helpful answers.

For example, the customer can ask:

- “Is this product suitable for babies aged 6 to 12 months?”
- “How often should this product be used?”

The system analyzes the question and provides an intelligent, helpful response based on the product information.

A “Back” button is available to return to the previous screen.

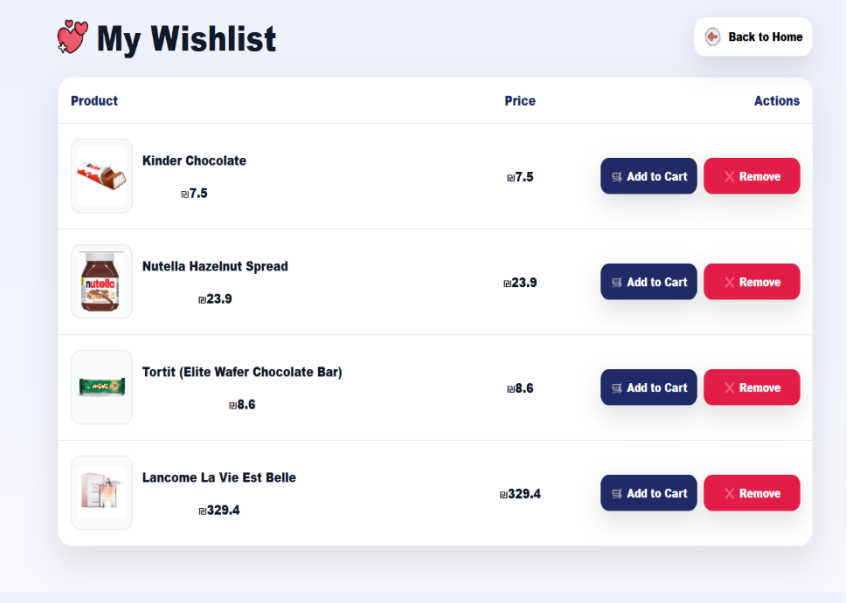
My Wishlist

Description:
This screen displays a list of products that the customer has added to their Wishlist.
For each product, the system shows the product image, product name, and price.

The customer can perform the following actions:

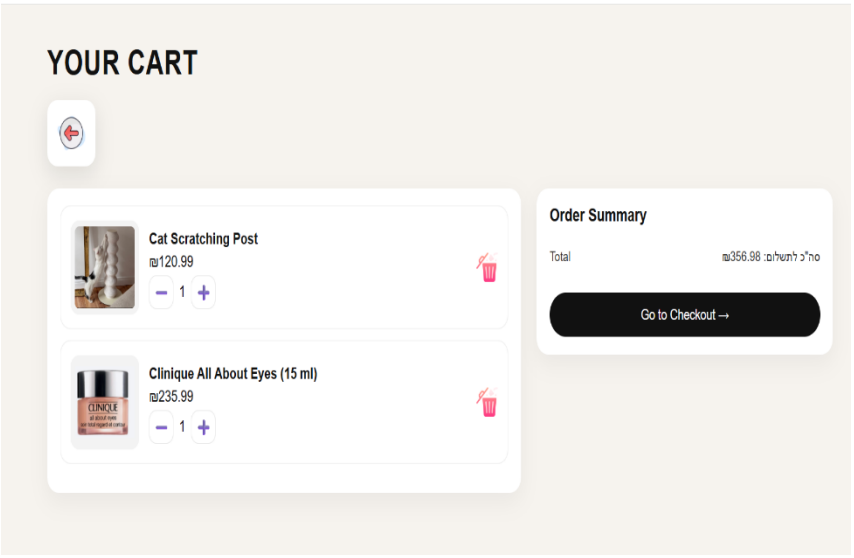
Add a product from the Wishlist to the shopping cart , Remove a product from the Wishlist.

The screen also includes a “Back to Home” button that allows the user to return to the main page of the application.



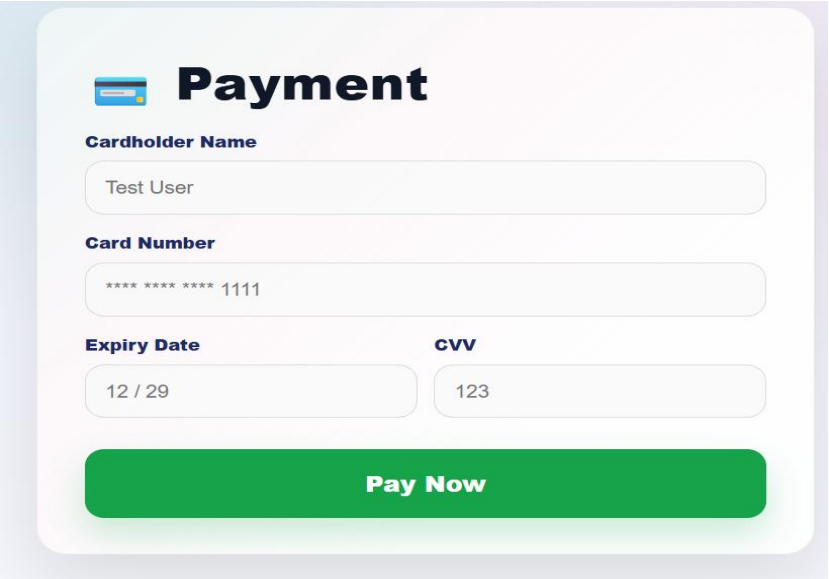
Screen: Shopping Cart

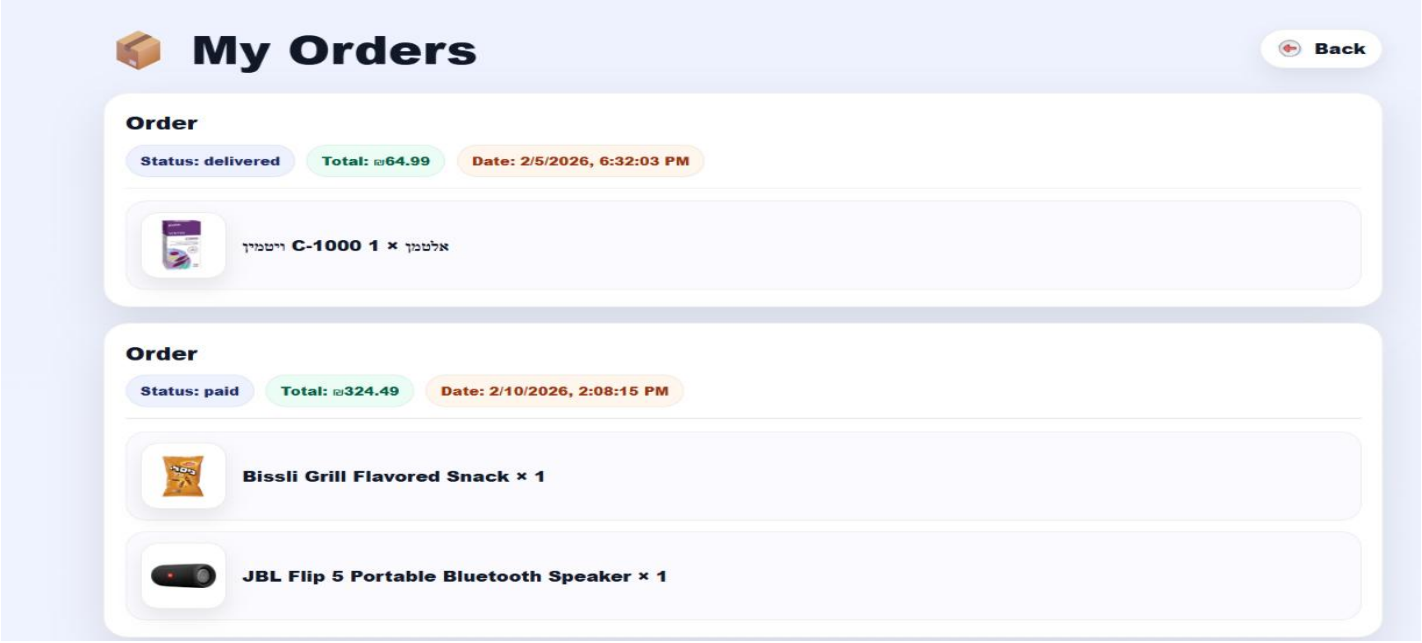
Description:
The screen displays all products added to the cart, including the product image, name, price, and quantity.
The customer can increase or decrease the quantity of each product, remove products from the cart, and view the total order price.
An order summary is shown on the side, and the customer can proceed to checkout by clicking the “Go to Checkout” button.



Screen: Payment

Description:
The customer enters the cardholder name, card number, expiry date, and CVV.
After filling in the required information, the customer clicks the “Pay Now” button to confirm the payment.
If the payment is successful, the order is created and processed by the system.





Screen: My Orders

Purpose:

The My Orders screen allows the customer to view all their past and current orders.

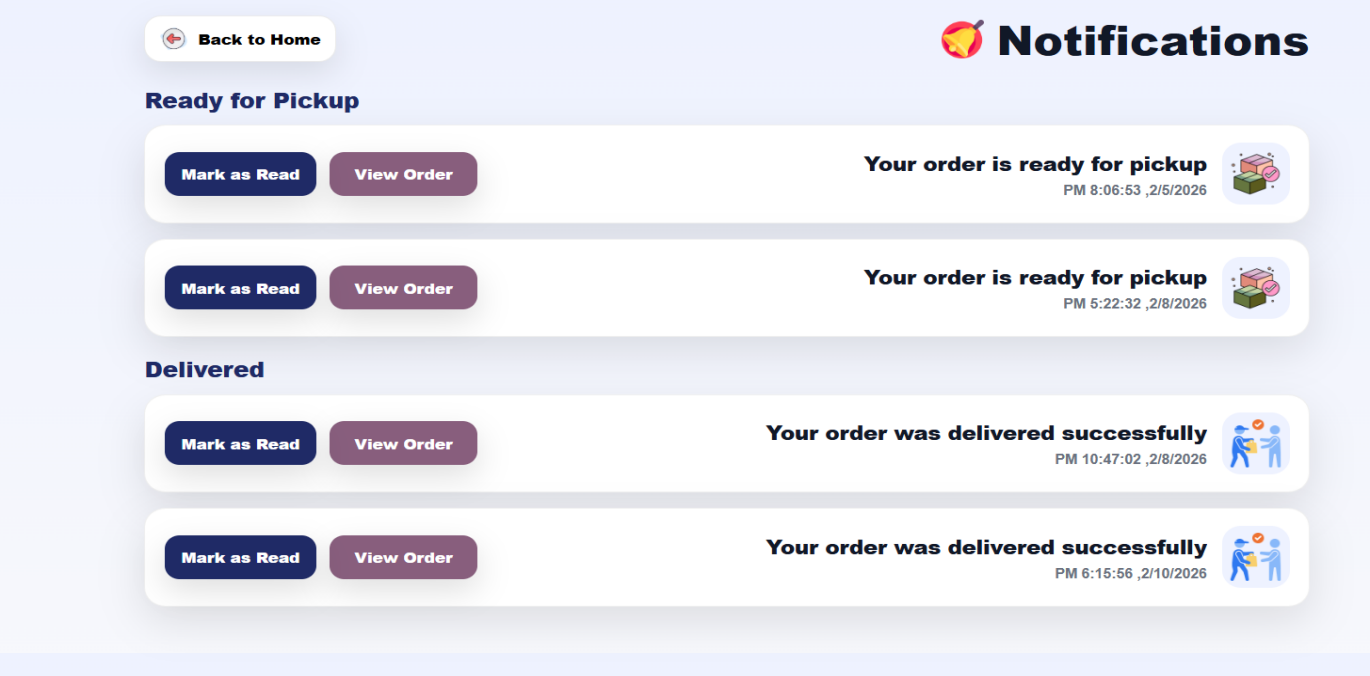
Description:

The screen displays a list of the customer's orders.

For each order, the system shows the order status (such as paid or delivered), the total price, and the order date.

Each order also shows the products included in it.

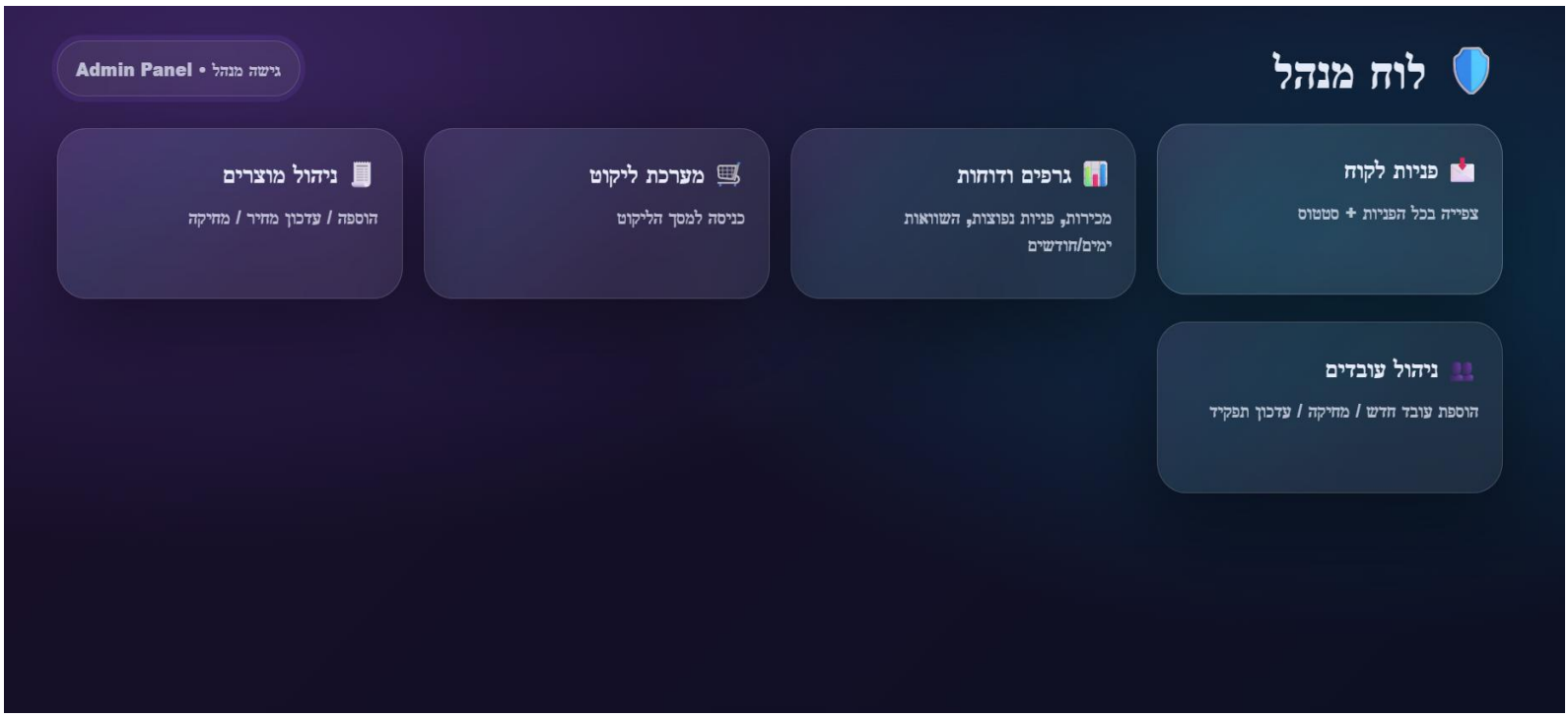
The customer can review their order history and track the status of each order.



Screen: Notifications

Purpose:
The Notifications screen informs the customer about important updates related to their orders.

Description:
The screen displays a list of notifications grouped by order status, such as *Ready for Pickup* and *Delivered*.
Each notification shows a message (for example: “Your order is ready for pickup” or “Your order was delivered successfully”) along with the date and time.
The customer can mark notifications as read or click **View Order** to see the order details.



Admin Panel Dashboard

The Admin Panel is the main control center for the system administrator. From this screen, the administrator can access all management modules of the system.

The dashboard contains several main sections:

- **Product Management** – Allows the admin to add new products, update existing products, change prices, and delete products from the system.
- **Order Picking System** – Provides access to the picking system, where the admin can view and monitor orders and their statuses.
- **Reports and Analytics** – Displays graphs and statistics such as sales data, top products, complaints statistics, and revenue reports.
- **Customer Requests** – Allows the admin to view and manage customer complaints and requests, and track their handling status.
- **Employee Management** – Enables the admin to add new workers, update worker details, and remove workers from the system.

This screen centralizes all administrative operations and provides the administrator with full control over the system.



Reports and Analytics Dashboard

This screen provides the administrator with a visual overview of the system's performance and customer activity using charts and graphs.

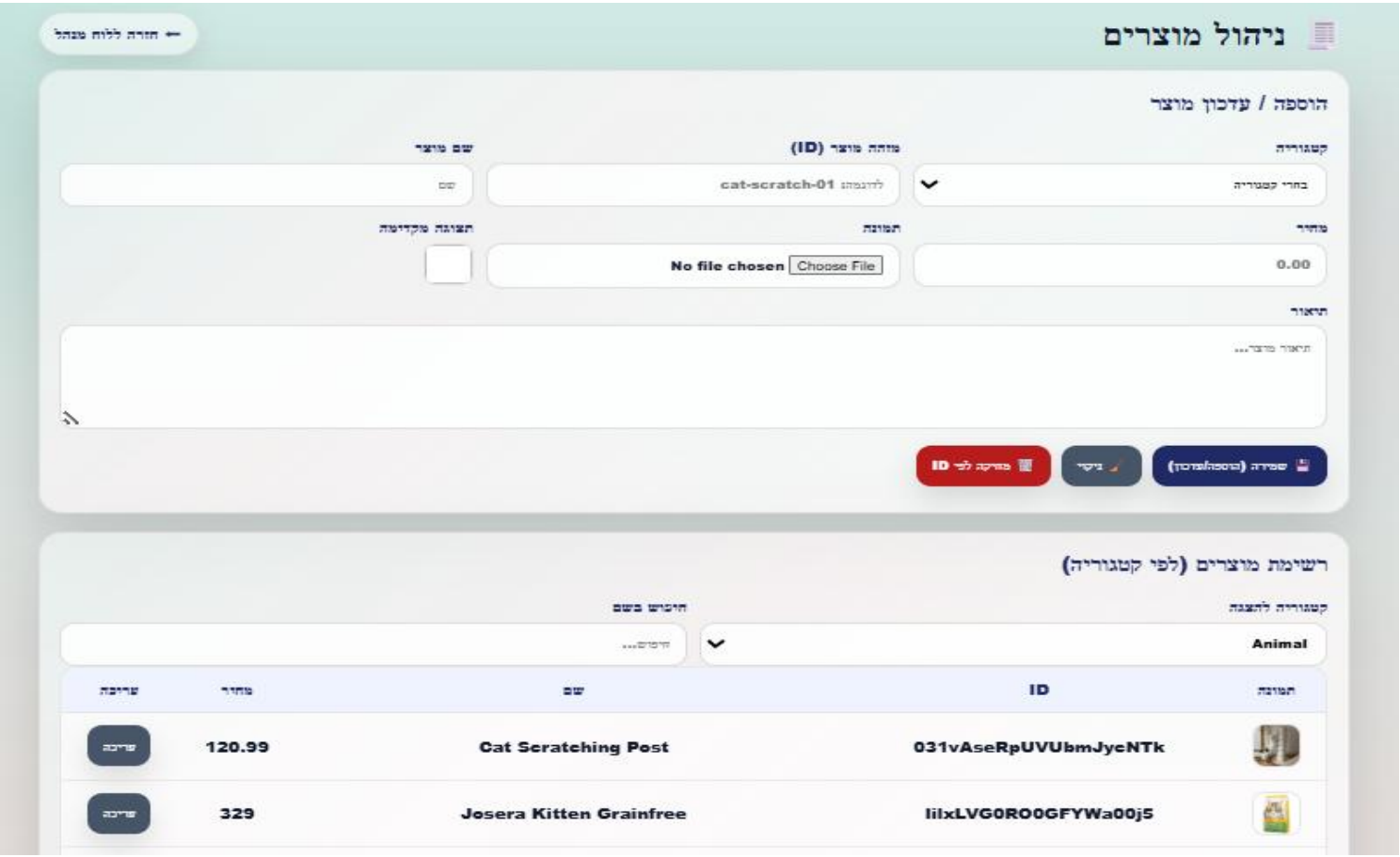
The dashboard includes:

- A chart showing customer request reasons (e.g., late delivery, missing item, damaged product, payment issues, bad service).
- A chart displaying the top-selling products.
- A revenue chart that shows the total income over time (by dates).

Using this screen, the administrator can:

- Monitor sales performance.
- Analyze customer complaints and common issues.
- Identify the most popular products.
- Track revenue trends over time.

This dashboard helps management make data-driven decisions, improve service quality, and optimize product and sales strategies.



Product Management Screen

This screen allows the administrator to manage the products in the system.

Using this screen, the administrator can:

- Add a new product by entering product details such as name, product ID, category, price, image, and description.
- Update an existing product's information, including its name, price, category, image, and description.
- Delete a product from the system.
- View the list of all products, displayed in a table with product image, ID, name, and price.
- Filter or search products by category or name to quickly find specific items.

This screen provides full control over the product catalog and ensures that the store's product information is always up to date.

14.Summary – Teamwork, Challenges, and Lessons Learned

This project was developed by two team members, Esraa and Laren. From the beginning, we divided the responsibilities between us and worked in parallel on different parts of the system. While this approach helped us progress faster, it also created challenges related to coordination, time management, and synchronization between different system components.

One of the main challenges was coordinating time between team members and working simultaneously on multiple parts of the system. We also had to deal with data synchronization between the customer side, the worker side, and the admin side, which required careful planning of the database structure and system logic.

During development and testing, we encountered system crashes and bugs, and we had to handle errors and inconsistent data in Firebase. Another important challenge was implementing different permissions and access levels according to the user type (Customer, Worker, Admin), and making sure that each user could access only the relevant features.

In addition, integrating multiple system modules (client, worker, admin, notifications, and AI assistant) required continuous communication and coordination between us.

Despite these challenges, the project helped us improve our technical skills, teamwork, problem-solving abilities, and understanding of how to build a complete real-world system.

Project Phases and Task Distribution

Month 1 – Analysis and Design

Both Esraa and Laren:

- Choosing the project topic
- Defining system goals
- Requirements analysis
- Defining user types (Customer / Worker / Admin)
- Designing screens and system flow
- Planning system architecture
- Designing the database structure (Firebase)
- Preparing Use Case and Activity diagrams

Month 2 – Infrastructure and Client-Side Development

Esraa:

- Developing the client side (Login/Signup, Home page, Categories, Product page, Cart, Favorites, My Orders, Customer Requests)
- Integrating AI for smart search and product questions

Laren:

- Setting up Firebase
- Designing and implementing the database structure

- Connecting the system to the database
- Configuring Authentication and basic security rules

Month 3 – Worker and Admin Development

Esraa:

- Developing the admin side (Product Management, Viewing Customer Requests, Analytics Dashboard, Admin Permissions)

Laren:

- Developing the worker side (Viewing Orders, Picking System, Updating Order Status)

Both:

- Integration testing between all system components

Month 4 – Testing, Documentation, and Submission

Both Esraa and Laren:

- Writing a Test Plan (Black Box)
- System testing and bug fixing
- Preparing the project book (Class, Sequence, DB, Architecture diagrams + explanations)
- Screen documentation
- Recording a demo video
- Packaging the code and preparing the final submission